

The chemistry that does work in the body

These notes follow the lecture slide by slide. Each section names the slides it covers, so you can read it before the video, follow it during the video, or use it after to check what you remember. The six solved problems from the slides are written out in full here. Silverthorn chapter 2.

COMPETENCIES 1 TO 4 OF WEEK 2

38 SLIDES · 6 SOLVED PROBLEMS

HOW TO USE THIS SHEET

Pass one, before the video: read the section, then close the page and put what you can into the box for that competency on your Competency Study Guide. Pass two, after the video: come back in a different color and add what you missed. Try each solved problem yourself before you read the work. Getting it wrong first is how the method sticks.

The problem for today

SLIDES 1 TO 2 OF THE DECK

Last week you drew a control loop, and every box in it was a molecule doing a job. The sensor was a protein that changes shape when something binds it. The signal was a molecule that fits one receptor and nothing else. The effector was a protein that turns binding into work. So the question for this week is what gives a molecule its job, and the short answer is shape. The long answer is this lecture.

Here is the problem we are going to solve by the end. A patient's arterial pH falls from 7.40 to 7.10. Nothing in their DNA has changed and no enzyme has been destroyed. Yet enzymes all over the body have stopped working properly. Before you know any chemistry, make a guess: did the hydrogen ion concentration rise by about 4 percent, by about 30 percent, or did it roughly double? Write your guess down. Solved problem 4 answers it.

Atoms, electrons and ions

SLIDES 3 TO 4 OF THE DECK

An atom has three particles and only one of them does chemistry. Protons decide which element you have. Neutrons decide which isotope of that element you have. Electrons do everything else. Every bond, every ion, every stored packet of energy and every free radical in the body is a story about electrons.

Electrons have four jobs in physiology. Shared between two atoms they make a covalent bond. Gained or lost they make an ion. Pushed up into a higher orbital they hold energy, which is how ATP and NADH carry it around. Left unpaired they make a free radical, which is reactive and does damage.

An isotope has the same number of protons and a different number of neutrons. Chemically it behaves the same as the

ordinary element, which is exactly why nuclear medicine works: the body takes up a radioactive isotope of iodine as if it were ordinary iodine, and the camera sees where it went. An ion is different. It has gained or lost electrons and now carries a charge, and charge changes what it can bind and where it can go. A hydrogen atom that loses its one electron is a bare proton, and that bare proton is the H^+ that pH measures.

Atomic number

The number of protons. It decides which element the atom is.

Isotope

Same element, different number of neutrons. Same chemistry, different mass. Some isotopes are unstable and give off radiation.

Ion

An atom or molecule that has gained or lost electrons and carries a net charge.

Cation

A positive ion. It lost electrons. Na^+ , K^+ , Ca^{2+} , H^+ , Mg^{2+} .

Anion

A negative ion. It gained electrons. Cl^- , HCO_3^- , phosphate, sulfate.

Electrolyte

A substance that separates into ions in water, so the solution conducts electricity.

Free radical

A molecule with an unpaired electron. Unstable and reactive. Relevant later to cell injury and to how radiation damages tissue.

Trace element

An element needed in tiny amounts, usually as a cofactor for a protein. Iron, iodine, zinc, copper.

SLIDE 7 OF THE DECK

The ions on the short list below run most of physiology. Sodium sets the volume of the fluid outside cells. Potassium sets the resting voltage of every cell and decides whether the heart can fire. Calcium triggers contraction and secretion. Chloride is the main negative ion outside cells. Bicarbonate is the main buffer in blood. And hydrogen ion, present at a millionth the concentration of sodium, sets the shape of every protein in the body.

One fact from this section generates most of weeks 3, 4 and 5, so say it back to yourself now: a charged particle cannot cross the oily middle of a membrane on its own. Every ion that moves in the body moves through a protein.

THE IONS TO KNOW NOW, AND WHERE EACH ONE SHOWS UP

ION	TYPICAL PLASMA VALUE	WHERE IT MATTERS
Sodium, Na⁺	135 to 145 mM	Extracellular fluid volume, membrane potential, most secondary active transport
Potassium, K⁺	3.5 to 5.0 mM	Resting membrane potential, cardiac excitability
Calcium, Ca²⁺	2.2 to 2.6 mM total	Muscle contraction, secretion, intracellular signaling
Chloride, Cl⁻	98 to 107 mM	Main extracellular anion, inhibition in neurons, stomach acid
Bicarbonate, HCO₃⁻	22 to 26 mM	The principal buffer in blood
Hydrogen, H⁺	about 40 nM	Sets pH, and therefore the shape of every protein

NURSING

Potassium matters more than the other electrolytes because the heart's ability to fire depends on it.

Example: Potassium is replaced with monitoring and never pushed fast, because the margin between correcting it and causing an arrhythmia is small.

RADIOLOGIC TECHNOLOGY

Ionizing radiation is named for exactly this. The beam knocks electrons off atoms in tissue and leaves charged particles behind.

Example: Most of that happens in water, because water is most of the tissue. The reactive fragments go on to damage DNA, and that chain is the reason dose limits and shielding exist.

MEDICINE

A charged particle cannot get through a membrane by itself. Every ion that moves needs a protein to go through.

Example: A drug that blocks one kind of channel can change the behavior of an entire organ.

RESPIRATORY THERAPY

Electrolytes decide how hard a muscle can pull, and breathing is muscle work.

Example: Low potassium or low phosphate weakens the diaphragm, which shows up as a failed weaning trial rather than as breathlessness.

A covalent bond is two atoms sharing a pair of electrons. It is strong, it takes real energy to make or break, and it is what holds a molecule together. If the two atoms share evenly the molecule is nonpolar, and molecules made mostly of carbon and hydrogen are like that. If one atom pulls the electrons harder the molecule is polar. Water is the polar molecule that matters. Its oxygen pulls the shared electrons toward itself, so the oxygen end is slightly negative and the two hydrogen ends are slightly positive.

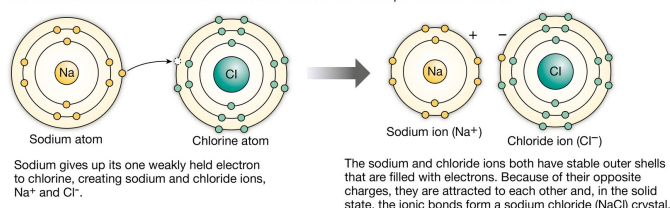
The weak bonds are the ones physiology actually runs on. An ionic bond is the attraction between opposite charges. A hydrogen bond is a slightly positive hydrogen on one polar molecule being drawn to a slightly negative oxygen, nitrogen or fluorine on another. Van der Waals forces are fleeting attractions between any two atoms that get close enough. Each of these is feeble on its own. In bulk they give water its surface tension, hold the two strands of DNA together, and fold every protein in your body.

Here is why the weakness is the point. A hormone that bound its receptor with a covalent bond would never let go, and the signal could never be switched off. Every reversible interaction in the body, binding, recognition, folding and unfolding, uses weak bonds because weak bonds can be undone.

Noncovalent Bonds

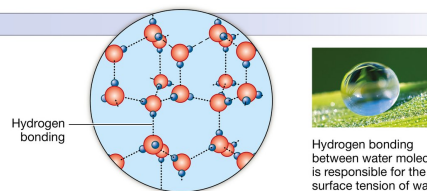
(c) Ionic Bonds

Ionic bonds are electrostatic attractions between ions. A common example is sodium chloride.



(d) Hydrogen Bonds

Hydrogen bonds form between a hydrogen atom and a nearby oxygen, nitrogen, or fluorine atom. So, for example, the polar regions of adjacent water molecules allow them to form hydrogen bonds with one another.



(e) Van der Waals Forces

Van der Waals forces are weak, nonspecific attractions between atoms.

Figure 2.6c to e. Ionic bonds, hydrogen bonds and van der Waals forces. Silverthorn 9e, Pearson.

Covalent bond

Two atoms sharing a pair of electrons. Strong. Single, double and triple bonds share one, two or three pairs.

Polar molecule

Electrons shared unevenly, so one region is slightly negative and another slightly positive. Water.

Nonpolar molecule

Electrons shared evenly, no partial charges. Mostly carbon and hydrogen. Fats.

Ionic bond

Attraction between a positive and a negative ion. Breaks easily in water.

Hydrogen bond

A weak attraction between a partially positive hydrogen and a partially negative O, N or F on another molecule.

Bonds: strong ones build, weak ones interact

SLIDES 5 TO 6 OF THE DECK

Van der Waals forces

Weak, nonspecific, short range attractions between atoms that are very close together.

Solved problem 1: the mass of a molecule

SLIDE 8 OF THE DECK

SOLVED PROBLEM 1

Glucose is $C_6H_{12}O_6$. Find its molecular mass, then do the same for table salt, NaCl.

Atomic masses: carbon 12, hydrogen 1, oxygen 16, sodium 23, chlorine 35.5, all in daltons. A dalton (Da) and an atomic mass unit (amu, also written u) are the same unit under two names, one twelfth the mass of a carbon 12 atom. Chemistry and physics say amu. Biology and medicine say dalton, especially for proteins, so this course uses Da.

THE WORK

- 1 Count each element in the formula. Glucose has 6 carbons, 12 hydrogens and 6 oxygens.
- 2 Multiply each count by its atomic mass. Carbon: $6 \times 12 = 72$. Hydrogen: $12 \times 1 = 12$. Oxygen: $6 \times 16 = 96$.
- 3 Add them. $72 + 12 + 96 = 180$ Da. One molecule of glucose weighs 180 daltons.
- 4 Sodium chloride: $23 + 35.5 = 58.5$ Da.
- 5 The same numbers in grams give you one mole. 180 g of glucose is one mole of glucose molecules, and 58.5 g of NaCl is one mole of NaCl. Keep both numbers. They come back in problems 2 and 3.

WHY MOLES MATTER

A mole is 6.02×10^{23} particles, and the molecular mass in grams is one mole of that substance. That is the bridge between weighing something, which a pharmacy can do, and counting molecules, which is what the body responds to. Equal masses of two different solutes are not equal numbers of molecules.

The four families of biological molecules

SLIDE 9 OF THE DECK

Every biological molecule belongs to one of four families: carbohydrates, lipids, proteins and nucleotides. Most contain carbon, hydrogen and oxygen. Proteins and nucleotides add nitrogen, and nucleotides and phospholipids add phosphorus. The families also combine. A protein carrying a lipid is a lipoprotein, a protein or lipid carrying a sugar is a glycoprotein or glycolipid, and those sugar-tagged molecules on the outside of the cell are how cells recognize one another. Your blood type is a glycolipid.

CARBOHYDRATES

SLIDES 10 TO 11 OF THE DECK

Sugars come single, paired and chained. The five carbon sugars ribose and deoxyribose build RNA and DNA. The six carbon sugars glucose, fructose and galactose are fuel, and glucose is the one the body regulates. Glucose and galactose are the same atoms

with one hydroxyl group pointed the other way, and that single difference means they need different enzymes and different transporters. That is the first place this week shape decides how the body treats a molecule.

Two sugars joined make a disaccharide. Sucrose is glucose plus fructose, maltose is glucose plus glucose, lactose is galactose plus glucose, and each needs its own enzyme to split it. Many adults stop making lactase. The lactose stays in the gut, pulls water in after it and feeds the bacteria of the colon. The problem is not the sugar, it is a missing protein whose binding site fits that one shape.

Long chains of glucose are polysaccharides. Animals store glucose as glycogen, plants store it as starch, and cellulose is glucose too, but joined by a bond humans cannot digest, which is why it counts as fiber.

LIPIDS

SLIDES 12 TO 13 OF THE DECK

A fatty acid is a chain of carbons with a carboxyl group at one end. No double bonds is saturated, one is monounsaturated, two or more is polyunsaturated. A double bond puts a kink in the chain, kinked chains cannot pack tightly, and that is why unsaturated fats are liquid at room temperature and saturated fats are solid. Glycerol plus one, two or three fatty acids gives a mono, di or triglyceride, and more than 90 percent of the body's lipid is triglyceride.

Lipids are mostly carbon and hydrogen, so they are nonpolar and do not dissolve in water. That single fact is the whole reason they need carrier proteins to travel in blood. Three lipid relatives matter for signaling and structure. Eicosanoids are modified 20 carbon fatty acids made on demand from the membrane and acting nearby; prostaglandins are the familiar example, and aspirin works by blocking the enzyme that makes them. Steroids all start from cholesterol: cortisol, aldosterone, estrogen and testosterone are cholesterol with small edits, and because they are lipids they pass straight through membranes. Phospholipids are two fatty acids plus a phosphate head. The head loves water and the tails hate it, so in water they arrange themselves into a double layer with the tails hidden inside. That double layer is next week.

PROTEINS

SLIDES 14 TO 16 OF THE DECK

Twenty amino acids build human proteins, and nine of them are essential, meaning we cannot make them and must eat them. Every amino acid has an amino group, a carboxyl group, a hydrogen and one variable R group on the same carbon. The R group is the only thing that differs between the twenty, and it is where the personality lives: some are charged, some are polar, some are oily, one contains sulfur. Which R groups sit where decides how the chain folds.

A peptide bond joins the amino group of one amino acid to the carboxyl group of the next, releasing a water molecule. It is covalent, so a chain does not fall apart on its own. Two to nine amino acids is an oligopeptide, ten to a hundred is a polypeptide, and more than a hundred is a protein. Insulin has 51, so you will see it called both.

Proteins have four levels of structure. Primary is the order of amino acids, set by the gene and held by peptide bonds. Secondary is local folding into alpha helices and beta sheets, held by hydrogen bonds along the backbone. Tertiary is the full three dimensional shape, held by weak bonds between R groups and sometimes by

a disulfide bridge between two cysteines. Fibrous proteins like collagen are insoluble; globular proteins dissolve. Quaternary is several folded chains held together by weak bonds, and hemoglobin is four.

Notice that only the primary structure is held by strong bonds. Everything above it is held by the weak bonds from earlier, and that is deliberate, because it lets a protein flex when something binds it. It also means heat, pH and salt can undo the fold without touching the chain. That is denaturation: the same amino acids in the wrong shape, with no function. Change the pH and you change which R groups carry a charge; change the charges and you change which ionic bonds hold. That is the mechanism behind the opening problem.

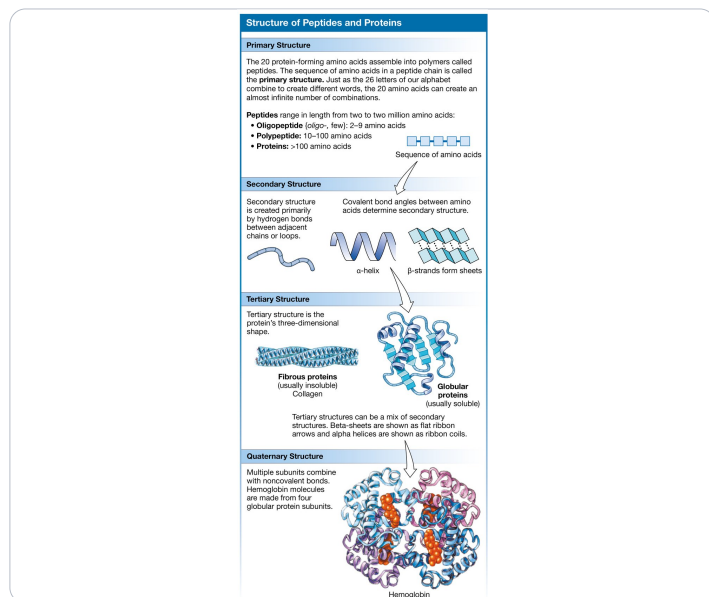


Figure 2.3. The four levels of protein structure, primary through quaternary. Silverthorn 9e, Pearson.

NUCLEOTIDES

SLIDE 17 OF THE DECK

A nucleotide is one or more phosphate groups, a five carbon sugar and a nitrogen containing base. Purines, adenine and guanine, have two rings. Pyrimidines, cytosine, thymine and uracil, have one. Nucleotides do two jobs. Single nucleotides catch energy in high energy electrons or phosphate bonds and hand it on: ATP, ADP, NAD and FAD. cAMP carries messages inside cells. Chains of nucleotides carry information as DNA and RNA. In DNA the bases pair by hydrogen bonds, A with T through two bonds and G with C through three, and a purine always pairs with a pyrimidine so the two strands stay the same distance apart.

FUNCTIONAL GROUPS

SLIDE 18 OF THE DECK

A functional group is a small cluster of atoms that shows up over and over and moves between molecules as a unit. The amino group, NH_2 , accepts H^+ and so acts as a base. The carboxyl group, COOH , gives up H^+ and acts as an acid. The hydroxyl group, OH , is polar and hydrogen bonds with water, which is why sugars dissolve. The phosphate group is charged and holds energy in ATP.

Phosphate is also the on switch. Adding a phosphate to a protein adds two negative charges at that spot, the nearby R groups shift, the fold changes and the binding site changes shape. Kinases add phosphate and phosphatases remove it, and that pair turns

proteins on and off in a huge fraction of cell signaling. You will meet it every week from here on.

Monosaccharide

A single sugar. Glucose, fructose, galactose, ribose, deoxyribose.

Glycogen

The storage form of glucose in animals, mostly in liver and muscle.

Triglyceride

Glycerol plus three fatty acids. The main storage form of fat.

Phospholipid

Two fatty acids plus a phosphate head. Amphipathic, and the building block of every membrane.

Steroid

A lipid built on four carbon rings, all derived from cholesterol.

Peptide bond

The covalent bond joining amino acids, formed with the loss of a water molecule.

Denaturation

Loss of a protein's shape, and therefore its function, by heat, pH or salt, without breaking the peptide bonds.

Disulfide bond

A covalent sulfur to sulfur link between two cysteines that helps hold a fold together.

Purine and pyrimidine

Two ring and one ring nitrogenous bases. In DNA a purine always pairs with a pyrimidine.

Glycoprotein

A protein with a sugar attached. Cell surface labels and blood types are glycoproteins and glycolipids.

Water, solutions and saying how much

SLIDES 19 TO 21 OF THE DECK

A solute is what dissolves, a solvent is what it dissolves in, and in the body the solvent is always water, so every body fluid is an aqueous solution. Hydrophilic molecules are polar or charged. Water surrounds them with a shell of its own molecules, and that hydration shell is what dissolving actually is. Hydrophobic molecules are nonpolar. Water has nothing to grab, so it pushes them together, and that is why oil beads up and why phospholipids line up with their heads in the water and their tails away from it.

Make the prediction from the slide before reading on. Two drugs are the same size, one hydrophilic and one hydrophobic. The hydrophobic one crosses a cell membrane easily, because the middle of the membrane is oily, but it needs a carrier protein to travel in blood. The hydrophilic one dissolves in plasma happily and crosses membranes with difficulty. Before you know anything else about a molecule, its polarity tells you where it can go and how it has to get there.

Molarity (M) = moles of solute ÷ liters of solution

Units: mol/L, written M. Body fluids are dilute, so values are usually mmol/L, written mM.

What it says: **molarity counts particles, not mass.** Equal masses of two solutes are not equal numbers of molecules, and it is the number of molecules that does the work.

Percent solution = grams of solute per 100 mL of solution

Units: g/100 mL. A 5 percent solution is 5 g in every 100 mL. Pharmacies and IV bags use this because they weigh solute, they do not count moles.

What it says: **the same percent means different numbers of particles for different solutes**, because their molecules weigh different amounts.

Equivalents (Eq) = moles × charge on the ion

Units: mEq/L for ions in blood. Na⁺ has 1 Eq per mole. Ca²⁺ has 2. HPO₄²⁻ has 2.

What it says: **one mole of calcium carries twice the charge of one mole of sodium.** Body fluids must stay electrically neutral, so charge matters as much as particle count.

Two more units you will read on lab reports. A deciliter, dL, is 100 mL, and glucose, calcium and cholesterol are usually reported in mg/dL in the United States. And whenever a value looks strange, check the unit before you check the physiology. Solved problem 3 shows why.

Solute

What dissolves.

Solvent

What it dissolves in. In the body, water.

Aqueous

Water based. Every body fluid.

Hydrophilic

Dissolves readily in water. Polar or charged.

Hydrophobic

Does not dissolve in water. Nonpolar.

Amphipathic

One polar region and one nonpolar region in the same molecule. Phospholipids.

Hydration shell

The layer of water molecules that surrounds a dissolved ion or polar molecule.

Concentration

Amount of solute per volume of solution, in whatever unit the report uses.

SOLVED PROBLEM 2

D5W is 5 percent dextrose (glucose) in water. (A) How many grams of glucose are in a 500 mL bag? (B) What is the molarity of D5W in mM? Glucose is 180 g per mole.

THE WORK

- Part A. A 5 percent solution is 5 g in every 100 mL. A 500 mL bag is five of those. $5 \times 5 \text{ g} = \mathbf{25 \text{ g of glucose}}$, with water added until the total volume is 500 mL.
- Part B. Work with one liter. 5 g per 100 mL is 50 g per 1000 mL, so 50 g/L.
- Convert grams to moles. $50 \text{ g} \div 180 \text{ g/mol} = 0.278 \text{ mol}$.
- That is 0.278 mol in one liter, so **0.278 M, which is 278 mM**.
- Why it matters: body fluids sit near 280 to 300 milliosmoles per liter, and D5W was designed to land close to that so the bag does not shrink or swell the red cells it meets. The pharmacy label reads 252 mOsm/L rather than 278 because the dextrose in the bag is the monohydrate form, 198 g per mole. Same reasoning, slightly heavier molecule. Next week you will see what happens once the glucose is taken up and only water is left.

SOLVED PROBLEM 3

Normal saline is 0.9 percent NaCl. NaCl is 58.5 g per mole and separates completely into Na⁺ and Cl⁻ in water. (A) Sodium concentration in mM? (B) In mEq/L? (C) Plasma calcium is 2.5 mmol/L. Calcium is Ca²⁺ and weighs 40 g per mole. Give it in mEq/L and in mg/dL.

THE WORK

- Part A. 0.9 percent is 0.9 g per 100 mL, so 9 g per liter. $9 \text{ g} \div 58.5 \text{ g/mol} = 0.154 \text{ mol/L}$. Every NaCl gives one Na⁺, so **sodium is 154 mM**, and chloride is 154 mM as well.
- Part B. Sodium carries one charge, so equivalents equal moles. **154 mEq/L**. Plasma is about 140 mEq/L, so saline is a little saltier than you are.
- Part C, equivalents. Calcium carries two charges. $2.5 \text{ mmol/L} \times 2 = \mathbf{5 \text{ mEq/L}}$.
- Part C, mass. $2.5 \text{ mmol/L} \times 40 \text{ mg/mmol} = 100 \text{ mg/L}$. A deciliter is a tenth of a liter, so **10 mg/dL**, which is the number you see on a metabolic panel.
- The lesson: the same calcium reads 2.5, 5 or 10 depending on the unit. Check the unit before you check the physiology.

Solved problems 2 and 3: the IV bags**SLIDES 22 TO 23 OF THE DECK****pH: a narrow window, defended hard****SLIDES 24 TO 25 OF THE DECK**

An acid gives up H⁺ in water; the carboxyl group does this. A base takes H⁺ out of solution, either by binding it directly, as ammonia

does, or by releasing OH^- that combines with H^+ to make water, as sodium hydroxide does. pH is a way of writing the hydrogen ion concentration so the numbers are manageable. Pure water has $1 \times 10^{-7} \text{ M}$ of H^+ , which is pH 7.

The scale runs 0 to 14 and each step is ten times the hydrogen ion concentration of the step above. That is the part people miss. From pH 8 to pH 6 is not twice the acid, it is a hundred times the acid. Arterial blood is held at 7.35 to 7.45. Inside most cells is a little lower, about 7.0 to 7.2. Stomach contents sit near 1 to 3, and urine can swing from about 4.5 to 8, because the kidney uses urine to dump whatever acid or base the blood needs to lose. A blood pH below 7.00 or above 7.70 is not compatible with life.

The body defends pH this hard because free H^+ can join in hydrogen bonds and ionic bonds. Enough of it changes which R groups on a protein carry a charge, the weak bonds holding the fold shift, the binding site changes shape and the protein works worse or not at all. A buffer is anything that soaks up H^+ when there is too much and releases it when there is too little. Proteins buffer, phosphate buffers, and bicarbonate is the big one in blood. You build the bicarbonate system properly in week 13.

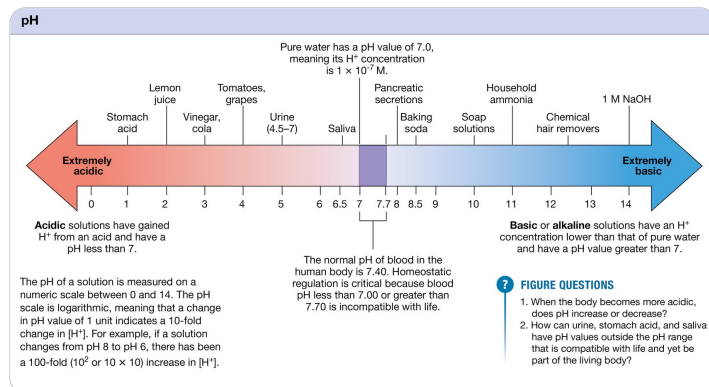


Figure 2.9. The pH scale, with body fluids and household examples placed on it. Silverthorn 9e, Pearson.

$\text{pH} = -\log [\text{H}^+]$

The log is the point. Square brackets mean concentration. A drop of one pH unit is a tenfold rise in hydrogen ion concentration, and a drop of 0.3 is a doubling.

What it says: **pH runs backwards.** More acid means a lower number, and small changes in the number are large changes in the acid.

$[\text{H}^+] = 10^{-\text{pH}}$

The same equation turned around. Use it to get from the pH on a blood gas report back to a concentration.

What it says: **normal arterial $[\text{H}^+]$ is about 40 nanomolar.** A millionth of the sodium concentration, and defended more tightly than anything else in the body.

SLIDE 26 OF THE DECK

SOLVED PROBLEM 4

Use $\text{pH} = -\log [\text{H}^+]$ and $[\text{H}^+] = 10^{-\text{pH}}$. (A) A solution has $[\text{H}^+] = 1 \times 10^{-7} \text{ M}$. What is its pH? (B) Normal arterial $[\text{H}^+]$ is 40 nM, which is $4 \times 10^{-8} \text{ M}$. Show that this gives pH 7.40. (C) Our patient dropped to pH 7.10. What is their $[\text{H}^+]$ now, and how many times higher is it than yesterday?

THE WORK

- 1 Part A. $\text{pH} = -\log(10^{-7})$. The log of 10 to a power is the power, so $\log(10^{-7}) = -7$, and the minus sign in front makes it **pH 7**.
- 2 Part B. Split 4×10^{-8} into two logs: $\log(4) + \log(10^{-8}) = 0.60 + (-8) = -7.40$. Flip the sign: **pH 7.40**. Worth memorizing: 40 nM is 7.40.
- 3 Part C. $[\text{H}^+] = 10^{-7.10}$. Write it as $10^{0.90} \times 10^{-8}$. $10^{0.90}$ is about 7.9, so $[\text{H}^+]$ is about **79 nM**.
- 4 $79 \div 40$ is close to **2**. The acid roughly doubled. A change of 0.30 on the pH scale is always a factor of 2, because $\log(2)$ is 0.30.
- 5 That answers the opening problem. The pH moved by a number that looks small, the hydrogen ion concentration doubled, and doubling the free protons is enough to change which R groups are charged on proteins throughout the body.

Acid

A molecule that releases H^+ in solution.

Base

A molecule that removes H^+ from solution, directly or by releasing OH^- .

pH

The negative log of the hydrogen ion concentration. Lower is more acid, and each unit is a factor of ten.

Buffer

A molecule or pair of molecules that binds H^+ when there is too much and releases it when there is too little, so pH changes less than it otherwise would.

Acidosis

Arterial pH below 7.35. Alkalosis is above 7.45. Both are week 13 and week 15.

Bicarbonate, HCO_3^-

The main buffer in blood, produced from carbon dioxide and water, and the link between the lungs and the kidneys in pH control.

NURSING

The first sign of an acid problem is often how the person is acting, not a number on a report.

Example: New confusion or agitation can arrive before anyone has drawn a blood gas.

RADIOLOGIC TECHNOLOGY

The body holds its pH in a very narrow window, so anything you inject has to be mixed to match it.

Example: Modern low osmolar contrast agents are tolerated better than the old ones because their chemistry sits closer to what the blood already is.

MEDICINE

A pH change that looks tiny is not tiny. Every whole step on the scale is ten times more acid.

Example: Falling from 7.4 to 7.1 roughly doubles the hydrogen ion concentration, which is enough to change protein shape throughout the body.

RESPIRATORY THERAPY

Reading acid and base status is the main skill of the entire specialty.

Example: Ventilation is the fast lever on pH, so a change in minute ventilation moves the pH within minutes, not hours.

Proteins at work: binding is the mechanism under all of it

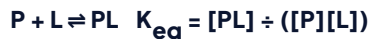
SLIDES 27 TO 29 OF THE DECK

Proteins do nine kinds of job: enzymes, membrane transporters, signal molecules, receptors, binding proteins, immunoglobulins, motor proteins, structural proteins and regulatory proteins.

Underneath, every one of them works the same way. Something binds, the protein changes shape, and the shape change does the job. The rest of the lecture is that one mechanism.

A ligand is anything that binds a protein: a substrate, a hormone, a drug, an ion. Specificity means a protein binds one ligand or a small family of very similar ones, because the binding site and the ligand have to be complementary in shape and charge. Neither side is rigid. Binding pulls the site into its working shape, which is called induced fit, and it is how binding can switch a protein on. Isoforms are closely related proteins that do the same job with different affinity, often in different tissues; fetal hemoglobin grips oxygen harder than the adult form. Two ligands that fit the same site compete for it. An agonist switches the protein on. An antagonist sits in the site and blocks it.

Affinity is how strongly a protein holds its ligand. Binding is reversible, so at any moment some pairs are forming and some are coming apart. At equilibrium the two rates are equal and the ratio of bound to unbound is constant. That ratio is the equilibrium constant, and the law of mass action says that if you push the concentration of anything on one side, the reaction runs toward the other side until the ratio is restored.



P free protein, L free ligand, PL the bound pair. K_{eq} is the equilibrium constant.

What it says: **a larger K_{eq} means a higher affinity**. More of the mix is sitting bound at any moment.

$$K_d = [P][L] \div [PL] = 1 \div K_{eq}$$

The dissociation constant. Numerically it is the ligand concentration at which half the protein is bound.

What it says: **a smaller K_d means a tighter grip**. A drug with a K_d of 1 nM works at very low concentration.

The **law of mass action** says that when protein binding is at equilibrium, the ratio of the bound and unbound components remains constant.

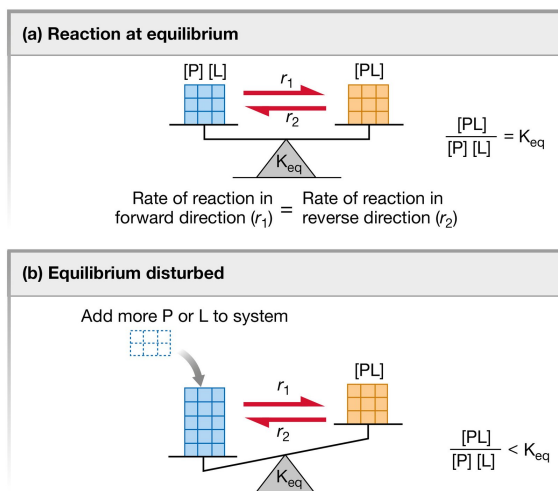


Figure 2.11a and b. At equilibrium the forward and reverse rates are equal. Add more protein or ligand and the ratio is disturbed, then restored. Silverthorn 9e, Pearson.

SLIDE 30 OF THE DECK

SOLVED PROBLEM 5

A drug binds receptor A with a K_d of 1 nM and receptor B with a K_d of 100 nM. Plasma drug concentration is 10 nM. (A) Which receptor has the higher affinity? (B) What fraction of each is occupied? Use fraction bound = $[L] \div ([L] + K_d)$. (C) The dose is raised so plasma hits 100 nM. What happens to each?

THE WORK

- 1 Part A. Smaller K_d is tighter binding. **Receptor A**, with a K_d of 1 nM, holds the drug 100 times more tightly than receptor B.
- 2 Part B, receptor A. $10 \div (10 + 1) = 10 \div 11 = 0.91$. About 91 percent of receptor A is occupied.
- 3 Part B, receptor B. $10 \div (10 + 100) = 10 \div 110 = 0.09$. About 9 percent of receptor B is occupied.
- 4 Part C. At 100 nM, receptor A: $100 \div 101 = 0.99$, essentially full and barely changed. Receptor B: $100 \div 200 = 0.50$, half full, up from 9 percent.
- 5 This is the logic of a therapeutic window. At the low dose the drug is almost entirely on its target. Raise the dose and the target gains almost nothing while the off target receptor starts to fill. Side effects arrive not because the drug changed, but because mass action moved.

Isoform

A closely related version of a protein with the same job and a different affinity, often in a different tissue.

Agonist and antagonist

An agonist binds and switches the protein on. An antagonist binds the same site and blocks it.

Law of mass action

Push the concentration on one side of a reversible reaction and it runs toward the other side until the equilibrium ratio is restored.

Turning proteins on, off, up and down

SLIDES 32 TO 33 OF THE DECK

There are three ways to switch a protein on. Cut it: proteolytic activation makes the protein in an inactive form and removes a piece to switch it on, which is how digestive enzymes and clotting factors work, and it is why a digestive enzyme does not digest the cell that made it. Give it a cofactor: the binding site does not work until an ion or a small organic molecule sits in it, and many trace minerals and B vitamins are cofactors. Or bind a modulator at a second site: the word allosteric means other site, and an allosteric activator binds away from the working site, shifts the fold and makes the site usable.

There are two ways to switch a protein off. A competitive inhibitor sits in the binding site itself, and it is beaten by adding more ligand because the two compete for the same spot. An allosteric inhibitor binds elsewhere and deforms the binding site, and more ligand does not help, because the site is the problem. Reversible modulators use weak bonds and let go. Irreversible ones bind covalently and the protein is done until the cell makes a new one. Aspirin inhibits its enzyme irreversibly, which is why one dose affects a platelet for the rest of that platelet's life.

Carbon monoxide is the competitive inhibitor to remember. It binds the same site on hemoglobin that oxygen does and holds it about 200 times more tightly. The treatment follows from mass action: flood the patient with 100 percent oxygen, sometimes at high pressure, so oxygen finally outnumbers the CO at the site.

SLIDES 34 TO 35 OF THE DECK

Warm a protein and its molecules collide more often, so activity rises. Keep warming and the weak bonds holding the fold start to break, and around 50 °C (122 °F) the protein in figure 2.13a has unfolded completely. pH does the same thing by a different route, by changing which R groups are charged. A mildly disturbed protein can refold when conditions return. A badly denatured one usually cannot. A cooked egg white does not become clear again. From the figure question: the protein is more active at 30 °C (86 °F) than at 48 °C (118 °F), because 48 °C is on the collapsing side of the curve, and a fever of 41 °C (105.8 °F) sits just past the peak, pulling activity down.

The cell also changes how many copies of a protein it has. With ligand held constant, rate rises in a straight line with the amount of protein, so up-regulation, making more copies or breaking down fewer, makes a cell more sensitive to a signal, and down-regulation does the reverse, over hours to days. With protein held constant, rate rises with ligand and then flattens. At saturation every binding site is busy, adding more ligand does nothing, and the only way to go faster is more protein.

SLIDE 31 OF THE DECK

PREDICT: HEMOGLOBIN AT THE MUSCLE

In the lung, hemoglobin sits in equilibrium with oxygen: $\text{Hb} + \text{O}_2 \rightleftharpoons \text{HbO}_2$. Blood now arrives at a working muscle where free oxygen is much lower. Using nothing but mass action, which way does the reaction run? Lower free oxygen means the ratio $[\text{HbO}_2] \div ([\text{Hb}][\text{O}_2])$ is now larger than K_{eq} , so the reaction runs in reverse. HbO_2 comes apart and releases oxygen until the ratio is restored. The muscle gets oxygen for no other reason than that it used some. No signal, no controller, just mass action. Week 13 adds why the curve is S shaped, but the direction is already decided here.

Ligand

Anything that binds to a protein.

Binding site

The region of a protein whose shape and charge are complementary to its ligand.

Specificity

A protein binding one ligand or a small family of closely related ones.

Induced fit

The binding site and ligand adjusting to each other as they meet, so binding itself changes the protein's shape.

Affinity

How strongly a protein holds its ligand. High affinity means a small K_d .

(a) Temperature and pH

Temperature and pH changes may disrupt protein structure and cause loss of function.

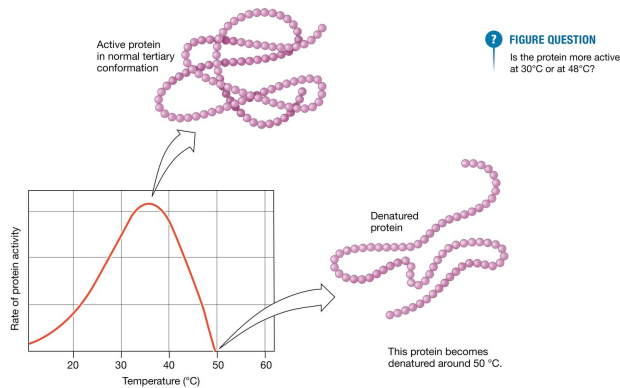


Figure 2.13a. Activity against temperature. It peaks near body temperature and collapses as the fold comes apart. Silverthorn 9e, Pearson.

(b) Amount of Protein

Reaction rate depends on the amount of protein.
The more protein present, the faster the rate.

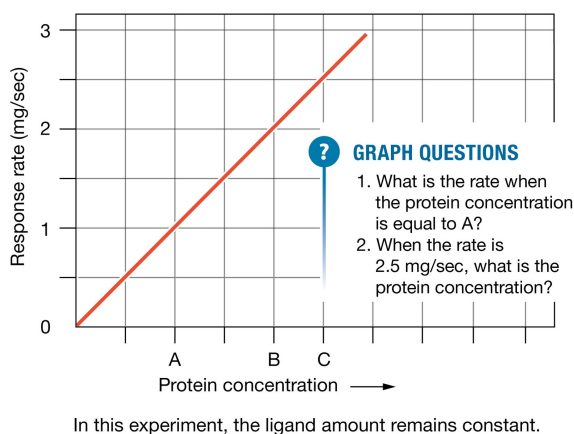


Figure 2.13b. Rate against amount of protein, ligand held constant. Silverthorn 9e, Pearson.

(c) Amount of Ligand

If the amount of binding protein is held constant, the reaction rate depends on the amount of ligand, up to the saturation point.

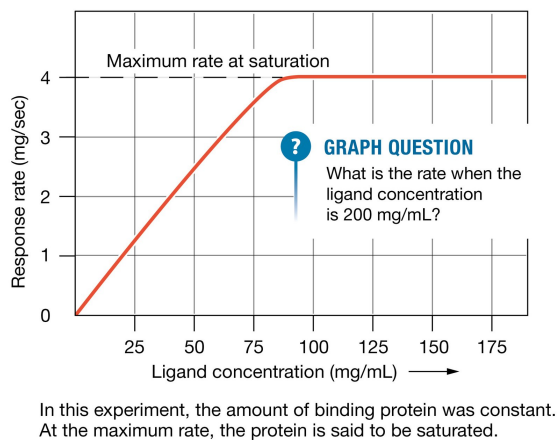


Figure 2.13c. Rate against amount of ligand, protein held constant. It flattens at saturation. Silverthorn 9e, Pearson.

SLIDE 36 OF THE DECK

SOLVED PROBLEM 6

Use figure 2.13b and c. (A) On the protein graph, what is the rate when protein concentration is A? At what protein concentration is the rate 2.5 mg/sec? (B) On the ligand graph, what is the rate at 200 mg/mL? (C) A patient's kidney reabsorbs glucose with a transporter. Blood glucose rises from 5 mM to 20 mM and glucose starts appearing in the urine. Which graph explains it, and what is the bottleneck?

THE WORK

- 1 Part A. At protein concentration A the line reads **1 mg/sec**. Follow 2.5 mg/sec across to the line and drop down: it lands at **C**. Rate is proportional to the amount of protein, so two and a half times the protein gives two and a half times the rate.
- 2 Part B. 200 mg/mL is past the bend. The curve flattened at about 100 mg/mL and stays at **4 mg/sec**. Doubling the ligand changed nothing, because every site was already busy.
- 3 Part C. This is the ligand graph. The kidney transporters are the protein and glucose is the ligand. At 5 mM they keep up and every glucose is reclaimed. At 20 mM they are saturated, the extra glucose has nowhere to bind, so it stays in the filtrate and leaves in the urine.
- 4 The bottleneck is the number of transporters, not the amount of glucose. Giving the kidney more time does not help. The only fix is on the protein side: fewer glucose molecules arriving, or more transporters made.
- 5 The rule: when a response stops increasing even though the input keeps rising, look for a saturated protein.

Proteolytic activation

Switching a protein on by cutting a piece off it.

Cofactor

An ion or small organic molecule that must sit in the binding site before the protein works.

Allosteric

Binding at a site other than the working site, changing the fold. Can activate or inhibit.

Competitive inhibitor

Blocks the binding site itself. Overcome by adding more ligand.

Up-regulation and down-regulation

A cell making more or fewer copies of a protein, changing how sensitive it is to a signal.

Saturation

Every binding site busy. Rate has hit its ceiling and only more protein raises it.

The patient, explained **start to finish**

SLIDE 37 OF THE DECK

Write the chain yourself before you read this one. Every arrow has to be something from this lecture.

- 1 pH fell by 0.30, so free H^+ roughly doubled, from about 40 nM to about 79 nM. Solved problem 4.
- 2 Extra H^+ binds to R groups that can accept it, amino groups especially. Charges that were negative or neutral become neutral or positive.
- 3 The ionic bonds and hydrogen bonds that hold the tertiary fold depend on those charges. Some break, some form where they should not.
- 4 The fold shifts, so the binding site shifts. Complementarity with the ligand drops, and with it affinity.
- 5 Lower affinity means less protein sits in the bound state at the same ligand concentration, so the rate of every reaction that protein runs falls. Mass action.
- 6 Now perturb it. The patient also has a fever of 40 °C (104 °F). On the curve in figure 2.13a, that is just past the peak and sliding toward denaturation, so the fever pushes on the same weak bonds and makes the shape problem worse, not better. Two perturbations, one mechanism.

SAY IT IN ONE SENTENCE

Everything a protein does depends on its shape, its shape is held by weak bonds, and weak bonds are at the mercy of temperature, pH and what is bound nearby. That is why the body defends its internal conditions as hard as it does.

What this week assumes you **already have**

If any line here is shaky, fix it before you work the week rather than after. The three review pages are linked from each card.

CHEMISTRY

Atoms, elements and the periodic table at the level of protons, neutrons and electrons
What a chemical formula like $C_6H_{12}O_6$ tells you
Acids and bases at the level of giving and taking H^+
What a log is, and that pH uses one

[Chemistry review page](#)

ANATOMY

The cell has a membrane, and a membrane separates inside from outside
Blood carries dissolved substances around the body

This is a physiology course, so anatomy shows up only where you need it to follow a mechanism. Two items is the whole list for this week, and that is not an oversight.

[Anatomy review page](#)

MATH

Metric prefixes: milli, micro, nano, deci
Percent, and moving between g/100 mL and g/L
Dividing by a molar mass to get moles
Reading a value off a graph and following it to the axis
Powers of ten and what a log does to them

[Math review page](#)

Close the page and **produce it**

These are the 4 competencies this lecture covers, straight off the study guide, with the two prompts that go with each one. Week 2 has three more, ATP and energy coupling, ATP production pathways, and the enzyme assay. Those are on the week 2 page and in the lab, not in this recording. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

01

Water and solution properties

Explain how the polarity and hydrogen bonding of water determine solubility and identify whether a given solute is hydrophilic or hydrophobic.

Do one of the two. Head your paper with the number.

1-1

Draw a water molecule and label the partial charges on it. Beside it draw one thing that dissolves in body water and one thing that does not, and write one sentence under each saying why. Then write what the body has to do for the one that does not dissolve.

1-2

Draw a blood vessel in cross section. Show one substance traveling free in the plasma and one traveling bound to a carrier protein. Label both, and write what decides which way a substance has to travel.

02

pH and buffers

Define pH, state the normal pH range of arterial blood, and explain how a buffer pair resists a change in pH when acid or base is added.

Do one of the two. Head your paper with the number.

2-1

Write the bicarbonate buffer equation from memory. Draw an arrow showing which way it shifts when acid is added and mark what happens to bicarbonate. Then write what the lung does to this equation and what the kidney does, and how fast each one works.

2-2

Draw the pH scale from 6.8 to 7.8. Mark the normal arterial range, mark which end is acidemia and which is alkalemia, and mark the survival limits. Under it write the equation for pH, and show with the numbers from solved problem 4 that 40 nM of hydrogen ion gives 7.40.

03

Protein structure and function

Relate the levels of protein structure to binding site shape and explain how denaturation by heat or pH change destroys function.

Do one of the two. Head your paper with the number.

3-1

Draw the four levels of protein structure as four small sketches in order and label each one. On the tertiary sketch mark where the binding site is, and write one sentence saying why the shape is the job rather than a consequence of the job.

3-2

Draw a folded protein with its binding site. Then draw it again after heat, and again after a pH change. Label what happened to the fold each time, and write what the protein can no longer do and whether it can recover.

04

Enzyme activity and regulation

Describe how enzymes lower activation energy and predict the effect of substrate concentration, temperature, pH, and competitive or allosteric inhibition on reaction rate.

Do one of the two. Head your paper with the number.

4-1

Draw rate against substrate concentration and label where saturation begins. Then draw the same axes twice more, once with a competitive inhibitor added and once with an allosteric one. Beside each, write what happened to the maximum rate and whether adding more substrate helps.

4-2

Draw rate against temperature, and rate against pH, on separate axes. Mark the optimum on each. Beside each curve write what is happening on the way up and what is happening on the way down, and say which side of the peak can be reversed and which cannot.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · The chemistry that does work in the body. Values given are typical teaching figures, not laboratory reference ranges. Figures © Pearson Education, Silverthorn Human Physiology 9e, used under the instructor license for this course.

Periodic table and molecular mass

Type a formula, or tap elements below to build one

Capital letter starts each element: NaCl, CaCl₂, Ca(OH)₂, C₆H₁₂O₆. Tapping an element adds one atom of it. Masses are in daltons, and the same number is grams per mole. The slides round to whole numbers (carbon 12, oxygen 16); this table keeps two decimals, so glucose reads 180.18 here and 180 on the slide.

Molecular mass

Nothing entered yet.

57 to 71 89 to 103

The four that make up 96 percent of the body: H, C, N, O The other elements physiology uses: major ions and trace elements Everything else

Cells and compartmentation

Read this before you watch anything. Fill your Competency Study Guide from here and from Silverthorn chapter 3 in your first color, then watch the concept videos and add the second color.

What this page covers

This is the structure half of Week 2. It follows Silverthorn chapter 3: how the body is divided into compartments, what the cell membrane is made of, what the organelles do, and how cells are organized into tissues and organs.

The other two thirds of Week 2, how things cross the membrane and how cells signal each other, are on their own pages. You need this one first, because both of those are about crossing or talking across the boundaries described here.

What you should already know

If any of these are shaky, fix them before you go further. Everything on this page leans on them.

- **Chemistry.** Polar and nonpolar molecules, hydrogen bonding, why oil and water separate, what a protein is made of, and what an ion is.
- **Anatomy.** The body cavities by name, and roughly where the major organs sit in them.
- **Math.** Percentages, and reading a value as a fraction of a whole. Volumes come up as fractions of body weight.

Chemistry review is on the optional review page. It carries no points and it is there whenever you need it.

1. Compartments, and why the body bothers

A is a functional space where conditions are kept different from the space next door.

Compartment: a space with its own conditions. It can be fully enclosed, like a cell, or only partly separated, like a functional zone inside a larger space. The boundary can be a membrane, a tissue layer, a bone, or nothing more than the fact that the two regions are kept chemically different.

Compartments cost the body something and buy it something, and you should be able to say both.

What compartments buy

- Processes that would interfere with each other get separated. Protein synthesis runs in one place, protein breakdown in another, and neither undoes the other.
- Two spaces sitting side by side can hold completely different contents.

- The extreme version is the lysosome, which holds enzymes at about pH 5 while the cytosol around it stays near pH 7. Rupture one and it damages the cell that made it.

What compartments cost

- Every barrier makes movement harder. Anything that has to get from one side to the other now needs a way through.
- That is the entire reason membrane transport exists as a topic, and it is why transport is the next thing you study this week.

Predict first

A cell keeps a digestive enzyme inside a membrane bag at pH 5. What two separate things has that bag done for the cell? Answer in your head before you open this.

– Check yourself

It gave the enzyme the acidic conditions it needs to work, and it kept that enzyme away from the cell's own proteins, which sit in a neutral cytosol where the enzyme is inactive anyway. Two jobs from one barrier: the right conditions in, and the damage contained.

2. The cavities and the fluid compartments

Three major body cavities

They are separated by bones and tissues and lined with tissue membranes.

CAVITY	BOUNDARIES	WHAT IT HOLDS
Cranial	The skull	The brain, the primary control center
Thoracic	Spine and ribs, with the diaphragm as the floor	The heart in its pericardial sac, and the lungs in separate pleural sacs
Abdominopelvic	Continuous abdomen and pelvis	Abdominal organs inside the peritoneum: stomach, intestines, liver, pancreas, gallbladder, spleen. The pelvic portion holds reproductive organs, the urinary bladder, and the terminal large intestine.

The kidneys are the exception worth memorizing. They sit , outside the abdominal cavity proper.

Retroperitoneal: behind the peritoneum. The organ lies between the peritoneal lining and the muscles and bones of the back, above the waist.

Other fluid filled compartments

- The circulatory system, vessels plus heart, is one large blood filled compartment.
- The eye is two: aqueous humor in front, vitreous humor behind.
- The central nervous system is bathed in cerebrospinal fluid around the brain and spinal cord.
- The pleural and pericardial sacs each hold a small volume of fluid.

The functional fluid compartments

This is the division that drives most of physiology.

- **ICF**, intracellular fluid, is the fluid inside cells.
- **ECF**, extracellular fluid, is everything outside cells.
- The cell membrane is the barrier between them.

ECF divides again into **plasma**, the fluid portion of blood, and **interstitial fluid**, the fluid that surrounds most body cells.

Predict first

A drug is injected into a vein. Name, in order, the compartments it has to pass through to reach the inside of a muscle cell.

– Check yourself

Plasma, then across the capillary wall into interstitial fluid, then across the cell membrane into intracellular fluid. Two barriers, three compartments. Most drugs that fail to work fail at one of those two crossings.

3. Lumens, the outside that runs through you

A is the interior space of a hollow organ. A blood vessel lumen contains blood.

Lumen: the open space inside a hollow tube or organ. It can be filled with air or with fluid.

Here is the idea that catches people out. Some lumens count functionally as the outside world, even though they run through the middle of your body. Material in the lumen of the digestive tract is not part of your internal environment until it crosses the wall of the gut. Think of a hole through a bead. The hole passes through the bead, but it was never inside it.

– What this means in the clinic

Nursing. E. coli living in the lumen of the large intestine is normal and harmless. Puncture the intestinal wall and the same organism is now in the internal environment, and that is a serious infection. Nothing about the bacterium changed. The compartment it sits in changed.

Rad tech. This is why oral contrast outlines the gut lumen so cleanly. The contrast stays in the tube, so what you see on the image is the shape of the outside world passing through the patient.

4. The cell membrane

Two different things are called a membrane

- A **tissue membrane** is a thin translucent layer of tissue lining a cavity or separating compartments. Mucous membranes, peritoneum, pleura and pericardium are all tissue membranes.
- A **cell membrane** is the phospholipid and protein boundary that separates ICF from ECF.

Drawings cause the confusion. Tissue membranes are usually drawn as a single line, which makes them look like cell membranes. They are not built the same way at all.

Cell membrane, plasma membrane, and plasmalemma all mean the same structure. This course says cell membrane, to keep it away from blood plasma.

Four jobs the cell membrane does

1. **Physical isolation.** It is the barrier separating ICF from ECF.
2. **Regulation of exchange.** It controls what ions and nutrients enter, how wastes leave, and what products get secreted out into the extracellular space.
3. **Communication.** Membrane proteins recognize and respond to signals from outside, and a change in the membrane can change what the cell is doing.
4. **Structural support.** Membrane proteins anchor the cytoskeleton inside and form junctions outside, both cell to cell and cell to matrix, which is what holds tissues together.

What it is made of

Lipids, proteins, and a small amount of carbohydrate. The ratio of protein to lipid varies by membrane, and it tracks how much work that membrane does. The inner mitochondrial membrane is about 75 percent protein, because that is where the ATP producing enzymes sit.

How the model got to where it is

- 1920s: chemists found enough lipid in a membrane for a double layer.
- 1960s: electron microscopy suggested a butter sandwich, a lipid layer between two protein layers.
- 1970s: freeze fracture electron microscopy showed the real three dimensional arrangement.
- 1972: Singer and Nicolson proposed the fluid mosaic model, which is still the model you use.

The fluid mosaic model

- A phospholipid bilayer, with hydrophilic phosphate heads facing water on both sides and hydrophobic fatty acid tails facing inward.
- Proteins embedded in it like raisins in bread.
- Glycoproteins and glycolipids on the extracellular surface.
- Total thickness about 8 nm.

5. Membrane lipids, the hydrophobic barrier

Three lipid types build the barrier: phospholipids, sphingolipids, and cholesterol.

Phospholipids

- A glycerol backbone, two fatty acid tails, and a phosphate group head.
- The head is polar and hydrophilic. The tails are nonpolar and hydrophobic.
- Put them in water and they arrange themselves to hide the tails from it. Nobody instructs them. The arrangement is the lowest energy option available.

Three arrangements worth telling apart

STRUCTURE	BUILD	WHERE IT MATTERS
Micelle	Small droplet, single layer, tails pointing inward	Fat digestion and absorption
Liposome	Larger sphere with a bilayer wall and a watery core	The core can carry water soluble molecules, which makes liposomes useful for drug delivery, cosmetics, and gene therapy. Immunoliposomes carry antibodies that target specific cancer cells, which improves specificity and cuts side effects. A liposome-like structure is also the likely precursor of the first cell.
Phospholipid bilayer	A sheet, heads out on both faces, tails in	The cell membrane itself

Sphingolipids

- Also have fatty acid tails, with heads that can be phospholipid or glycolipid.
- Their tails are longer than phospholipid tails, which matters in a moment when lipid rafts come up.

Cholesterol

- Mostly hydrophobic, and it inserts itself between the phospholipid heads.
- It decreases permeability to small water soluble molecules.
- It keeps the membrane workable across a wide range of temperatures. At normal body temperature, 37 degrees C (98.6 degrees F), the membrane has to stay fluid enough for proteins to move and firm enough to hold together, and cholesterol is what keeps both true.

6. Membrane proteins

Membrane proteins may account for almost one third of all proteins coded in your DNA, and a single cell has somewhere between 10 and 50 different types in its membrane. That number alone tells you the membrane is doing most of the cell's business.

Integral and peripheral

- **Integral proteins** are tightly bound. Removing one means disrupting the membrane with detergents. They include transmembrane proteins and lipid anchored proteins.
- **Peripheral proteins** attach by noncovalent interactions to other membrane proteins and can be removed without destroying the membrane. They work as enzymes, structural binding sites, and cytoskeleton anchors.

Transmembrane proteins

- They cross the membrane completely, once or as many as twelve times. A lot of the important ones cross seven times, and you will meet that seven pass shape again when you study signaling.
- Extracellular loops can carry carbohydrate attachments. Intracellular loops can carry phosphate attachments.
- Adding or removing a phosphate group changes what the protein does. That is how a signal from outside becomes a change inside.
- They stay put because the membrane spanning stretch of amino acids is nonpolar and interacts strongly with the phospholipid tails.

Lipid anchored proteins

- Either covalently bound to lipid tails inserted in the bilayer, or held on the outer surface by a .
- Many associate with sphingolipids to form lipid rafts, which are specialized membrane patches. Because sphingolipid tails are longer, the raft sits slightly raised above its neighbors.

GPI anchor: glycosylphosphatidylinositol. A lipid joined to a sugar phosphate chain, which tethers a protein to the outer surface of the membrane without that protein ever crossing it.

Do the proteins actually move

The original fluid mosaic model said proteins drift laterally through the bilayer. The picture has been updated. Some integral proteins are anchored to the cytoskeleton and are effectively immobile, and that restriction is what makes cell polarity possible: different faces of the same cell carry different proteins and behave differently. Transporting epithelia depend on it completely, and you will see why when you study how the intestine and kidney move things in one direction.

7. Carbohydrates and the glycocalyx

- Membrane carbohydrates attach only on the extracellular surface. Sugars on proteins make glycoproteins, sugars on lipids make glycolipids.

- Together they form the , a protective sugar layer on the outside of the cell.
- Glycoproteins matter in immune recognition. Your immune system reads this layer to decide what is yours.
- ABO blood groups are determined by the number and composition of sugars attached to membrane sphingolipids. Blood type is a sugar pattern on a cell surface, nothing more exotic than that.

Glycocalyx: the carbohydrate layer on the outer surface of the cell membrane, made of the sugar chains of glycoproteins and glycolipids.

– What this means in the clinic

Nursing. A transfusion reaction is a mismatch read at this sugar layer. The recipient's antibodies recognize surface sugars that are not theirs, and the response follows from that recognition.

Rad tech. Surface recognition is also why targeted contrast and radiopharmaceutical agents work. They are built to bind something specific sitting on the outside of a particular cell type.

8. Inside the cell

Cells differ because genes are used differently

You have trillions of cells and over 200 cell types. Every one of them inherited the same DNA. is what makes them different, and the shape of a mature cell reflects the job it ended up with.

Differentiation: selective gene activation, so that a cell switches on some genes and not others, which changes its structure and its function.

How a typical animal cell is laid out

- The cell membrane separates ICF from ECF.
- Inside sits the cytoplasm and the nucleus.
- Cytoplasm includes the cytosol, inclusions, insoluble protein fibers, and the membrane bound organelles.

The four cytoplasm components

COMPONENT	WHAT IT IS
Cytosol	The intracellular fluid itself, a semi gel holding dissolved nutrients, proteins, ions and wastes.
Inclusions	Insoluble particles with no membrane, so they sit in direct contact with the cytosol. Stored nutrients appear here as glycogen granules and lipid droplets. Others are protein or RNA protein combinations.
Protein fibers	Insoluble polymers providing structural support and movement.
Organelles	Membrane bound compartments that isolate substances and separate functions.

The dividing line to hold on to is the membrane. Inclusions have none and touch the cytosol directly. Organelles have one or more and do not.

Ribosomes

- Inclusions made of RNA and protein.
- They build proteins under the direction of DNA.
- Free ribosomes float in the cytosol and can group into polyribosomes of about 10 to 20.
- Fixed ribosomes attach to an organelle surface.
- A ribosome can switch between free and fixed.
- Cells that export a lot of protein carry the most of them, which is a useful thing to notice on a slide.

9. The cytoskeleton and motor proteins

Three families of protein fiber

FIBER	SIZE AND PROTEIN	EXAMPLES
Microfilaments	Thinnest. Actin.	Cell shape and movement
Intermediate filaments	Middle. Various proteins.	Keratin in hair and skin, neurofilament in nerve cells
Microtubules	Largest, hollow. Tubulin.	Tracks for transport, cilia, flagella, centrioles

Many accessory proteins associate with these fibers. Broadly the fibers do two things: structural support, and movement when motor proteins work along them.

What the cytoskeleton is

A flexible, dynamic, three dimensional scaffold of actin microfilaments, intermediate filaments and microtubules. Some of it is permanent. Most of it is assembled and disassembled as the cell needs it, which is the part students tend to forget.

Five jobs

1. **Cell shape.** Mechanical strength, and support for structures like microvilli.
2. **Internal organization.** It stabilizes where organelles sit, while the whole arrangement stays dynamic.
3. **Intracellular transport.** It acts as the tracks that organelles and vesicles move along. In a neuron that can mean moving cargo up to a meter.
4. **Assembly into tissues.** It connects to extracellular fibers, links cells to each other and to the matrix, and allows information transfer from cell to cell.
5. **Movement.** White blood cells squeezing out of vessels, growing neurons extending processes, cilia and flagella beating. Motor proteins burn ATP to move along the fibers.

Motor proteins

A motor protein converts the energy in ATP into directed movement. Three groups:

- **Myosin** binds actin and is the key player in muscle contraction.
- **Kinesin** moves vesicles along microtubules.
- **Dynein** also moves vesicles along microtubules, and drives the bending of cilia and flagella.

Most motor proteins have multiple protein chains and three regions: two heads that bind the cytoskeleton fiber, a neck,

and a tail that binds the cargo. Movement happens because the heads alternate binding and releasing, stepping along the fiber using ATP.

Predict first

A drug blocks ATP production in a neuron but leaves the microtubules intact. What happens to cargo that needs to travel down the axon?

– Check yourself

It stops. The track is fine, but the stepping is what costs ATP, and the motor protein cannot alternate binding and releasing without it. Intact road, no fuel.

10. Centrioles, cilia and flagella

Centrosome and centrioles

- The centrosome is the microtubule organizing center. It assembles tubulin into microtubules and sits near the nucleus.
- Most animal cells have 2 centrioles per centrosome.
- Each centriole is a cylinder of 27 microtubules arranged as nine triplets.
- In cell division they direct the movement of DNA.
- Mature nerve cells lack centrioles, because they no longer divide.

Cilia

- Short hairlike projections. Most cells have one short cilium.
- Cells lining the upper airways and part of the female reproductive tract carry many, and their coordinated beating moves fluid or secretions along.
- The cilium membrane is continuous with the cell membrane.
- Motile cilia have a 9 plus 2 core: nine pairs of microtubules around a central pair, anchored by a basal body inside the cell.
- Movement comes from microtubule pairs sliding past each other, driven by dynein, which produces rhythmic beating.

Flagella

- Same microtubule arrangement as cilia, much longer.
- In humans only sperm has one, and it has exactly one.
- Wave like motion propels the cell through fluid. The mechanism is the same as cilia.

Primary cilia, the nonmotile ones

- Most body cells carry a single stationary primary cilium.
- Structurally it lacks the central pair, so it is 9 plus 0.
- Its job is sensing the external environment. Photoreceptors use one for light sensing, and kidney cells use one to sense fluid flow.

– What this means in the clinic

Nursing. Mutations in ciliary proteins cause the ciliopathies, a group that includes polycystic kidney disease, vision loss, and some cancers, with links to development and to obesity. One malfunctioning structure, effects in several organs, because the same structure is used in several places.

Rad tech. Polycystic kidney disease is a classic imaging finding, and knowing it comes from a ciliary defect explains why other organ findings often travel with it.

Respiratory therapy. Airway clearance depends on coordinated ciliary beating. When that beating is impaired, secretions stay put, which is exactly the problem clearance therapy is trying to solve.

11. The organelles

Organelles are separated from the cytosol by one or more phospholipid membranes. That membrane isolates harmful substances and keeps processes apart.

Mitochondria

- They produce most of the cell's ATP.
- A double membrane creates two compartments. Inside the inner membrane is the matrix, which holds enzymes, ribosomes, granules and mitochondrial DNA, so a mitochondrion can make some of its own proteins. Between the membranes is the intermembrane space, which is important in ATP production.
- Mitochondrial DNA has a different sequence from nuclear DNA.
- The explains why.
- They replicate by budding, independently of cell division. Exercise can increase the number of mitochondria in muscle.
- Mitochondrial number tracks energy demand. Skeletal muscle is high, adipose tissue is low.

Endosymbiont theory: mitochondria descended from bacteria that invaded ancient cells and stayed. The evidence is that mitochondrial DNA and RNA enzymes resemble those of bacteria, and that mitochondria carry their own DNA and replicate on their own schedule.

Endoplasmic reticulum

A network of interconnected membrane tubes with three jobs: synthesis, storage, and transport of biomolecules.

	ROUGH ER	SMOOTH ER
Ribosomes	Attached	None
Main work	The main site of protein synthesis. Proteins are inserted into the ER lumen and chemically modified there.	Lipid, steroid and fatty acid synthesis. Makes the phospholipids for membranes. Modifies cholesterol into steroid hormones such as estrogen and testosterone. Detoxifies drugs in liver and kidney.
Specialization		In skeletal muscle, a specialized smooth ER stores the calcium used for contraction.

Golgi apparatus

- Stacked curved sacs, called cisternae, plus the vesicles around them.
- It receives proteins from rough ER, modifies them, and packages them into vesicles.

Cytoplasmic vesicles

- **Secretory vesicles** hold proteins that will be released from the cell.
- **Storage vesicles** keep their contents in the cytoplasm.

Lysosomes

- Storage vesicles that act as the digestive system of the cell. Their enzymes break down bacteria and worn out organelles. Reusable molecules go back to the cytosol and the rest is expelled.
- Up to 50 enzyme types, depending on the cell.
- The safety mechanism is pH. A newly formed lysosome sits at about pH 7.0 to 7.3 and its enzymes are inactive. Over time it accumulates hydrogen ions, using energy to do it, and when the pH drops to about 4.8 to 5.0 the enzymes switch on.
- Enzymes are sometimes released outside the cell on purpose, for instance to dissolve extracellular material such as bone mineral.
- Lysosomes can digest the cell's own material, leading to loss of cell mass. Muscle atrophy and the shrinking of the uterus after pregnancy both work this way.

– What this means in the clinic

Nursing. In rheumatoid arthritis, joint tissue damage is linked to inappropriate release of lysosomal enzymes. In the lysosomal storage diseases, a specific enzyme is missing. Tay Sachs is the example to know: defective lysosomes fail to break down glycolipids, the glycolipids accumulate in neurons, and the child loses vision and coordination. It is fatal in early childhood.

Rad tech. Muscle atrophy on imaging is lysosomal digestion of the cell's own proteins, not cells disappearing. Understanding that changes how you read a follow up study.

Peroxisomes

- Smaller than lysosomes.
- They degrade long chain fatty acids and other toxic molecules.

- Doing so generates hydrogen peroxide, which is itself toxic, so catalase converts it to water and oxygen.
- Peroxisomal disorders disrupt lipid processing and impair neural function by altering nerve membrane structure.

12. The nucleus, and protein synthesis as a compartment story

Key parts

- The nucleus holds the DNA that controls cell processes.
- The nuclear envelope is a double membrane separating nucleus from cytoplasm, with pores through it.

Nuclear pore complexes

- Large protein complexes with a central channel.
- Small molecules and ions move freely when a pore is open.
- Large molecules, proteins and RNA require energy dependent transport.
- That transport is specific, and the specificity is what keeps DNA inside the nucleus and directs enzymes to the right compartments.

Chromatin and nucleoli

- Chromatin is DNA plus associated proteins, granular in appearance in nondividing cells.
- There are 1 to 4 nucleoli per nucleus, dark bodies of DNA and RNA protein, containing the genes and proteins for ribosomal RNA synthesis.

Protein synthesis, start to finish

Follow the compartments. Every numbered step is a compartment boundary or a modification that only happens because of one.

- 1**Transcription. RNA is made from a DNA template, inside the nucleus.
- 2**RNA exits the nucleus through the nuclear pores.
- 3**Translation on free ribosomes in the cytosol, or
- 4**Translation on ribosomes attached to rough ER.
- 5**The new protein enters the rough ER lumen for modification.
- 6**It is packaged into a vesicle.
- 7**The vesicle fuses with the Golgi, and the protein is modified further in the Golgi lumen.
- 8**It leaves the Golgi in storage vesicles, or
- 9**it leaves in secretory vesicles and is released to the extracellular fluid.

Predict first

A protein destined for secretion is built on a free ribosome in the cytosol by mistake. What does it miss out on?

– Check yourself

Every modification that happens inside a lumen. It never enters the rough ER, so it is not modified there, never reaches the Golgi for further modification, and is never packaged into a

secretory vesicle. It is stuck in the cytosol in an unfinished form. The compartment is not a detour on the way out, it is where the protein gets finished.

13. Tissues

Single cells cannot perform all body functions, so cells specialize and work in groups. A **tissue** is a group of similar cells plus the extracellular matrix around them, working toward a common function.

Cells are held together by cell junctions and by the extracellular matrix.

Tissue complexity varies. A simple tissue is one cell type, such as the lining of a blood vessel. A complex tissue has many cell types and extensive matrix, such as connective tissue.

Histology

describes tissues by cell shape and size, arrangement in layers or scattered, the type of connections between cells, and the amount of extracellular material.

Histology: the study of tissue structure and function.

The four primary tissue types

1. Epithelial
2. Connective
3. Muscle
4. Neural

14. The extracellular matrix

The **extracellular matrix** is material outside cells, synthesized and secreted by those cells. It is not inert. It actively regulates growth, development, differentiation and apoptosis, and abnormalities in it are linked to heart failure and to cancer metastasis.

Two components, always

- **Proteoglycans.** Glycoproteins, protein plus polysaccharide chains. They trap water, which is what determines how hydrated a tissue is.
- **Insoluble protein fibers.** Collagen gives tensile strength. Fibronectin and laminin anchor cells to the matrix.

Matrix to cell attachments connect the matrix to membrane proteins to the cytoskeleton, which allows outside-in signaling. A cell can be told something by the material it is sitting in.

How much matrix, and how stiff

- Very little matrix: muscle and nerve.
- Extensive matrix: cartilage, bone, blood.
- Consistency runs across a spectrum: watery in blood and lymph, gel like in cartilage, rigid in bone.

Blood being a connective tissue surprises people every year. Read that spectrum again and it stops being strange. The matrix is plasma, and it happens to be liquid.

15. Cell junctions

Cell adhesion molecules

- CAMs are membrane spanning proteins.
- They handle both transient adhesion and permanent junctions.
- The interactions are weak and reversible, which is exactly what lets cells move.
- Examples: NCAMs guide growing neurons, and CAMs make white blood cell migration and platelet adhesion possible.

Three functional categories

1. Communicating junctions, gap junctions

- Direct cytoplasm to cytoplasm communication.
- Built from proteins called connexins, forming channels that open and close.
- They pass ions, small molecules, and electrical signals.
- Common in cardiac and smooth muscle, liver, pancreas, and endocrine tissues.

2. Occluding junctions, tight junctions

- They restrict paracellular movement, which is movement between cells rather than through them.
- Built from claudins and occludins.
- The barrier is dynamic. It can become leakier or tighter.
- Key roles: the intestine and kidney use them to control absorption, and the blood brain barrier depends on them.
- The brick wall analogy is the useful one. The cells are the bricks and the tight junctions are the mortar.

3. Anchoring junctions

These provide mechanical strength, either cell to cell or cell to matrix.

ANCHORS	CAM	TYPES
Cell to cell	Cadherins	Adherens junctions link actin filaments. Desmosomes link intermediate filaments, have dense plaques, are the strongest junction, and come as spot or belt desmosomes.
Cell to matrix	Integrins	Hemidesmosomes anchor intermediate filaments to laminin. Focal adhesions link actin to fibronectin and are important in signaling and migration.

– What this means in the clinic

Nursing. A blister is anchoring junction failure. In pemphigus, an autoimmune attack on junction proteins produces the same separation without the friction. In cancer metastasis, cadherins decrease, proteases are secreted, and matrix metalloproteinases digest the extracellular matrix so cells can invade. A cell that lets go and can cut its way out is a cell that can travel.

Rad tech. The blood brain barrier is built on tight junctions, and that is why contrast enhancement on a

brain study means something. Enhancement says the barrier is disrupted at that spot.

infection rate. Both are small facts that explain a lot of clinical behavior.

16. Epithelial tissue

What it does and how it is built

- Protects the internal environment, regulates exchange, covers surfaces and lines cavities.
- Everything entering or leaving the body crosses an epithelium. That sentence is worth keeping.
- One or more cell layers, minimal extracellular matrix.
- A basal lamina, also called the basement membrane, of collagen, laminin and proteoglycans anchors the epithelium to the tissue underneath.

Leaky and tight

- **Leaky epithelia** use anchoring junctions and allow paracellular transport. Capillaries are the example.
- **Tight epithelia** use tight junctions and force transcellular transport, through the cells. Kidney is the example.
- Tightness equals selectivity. The intestine can modulate its own tightness, which means the same tissue can change how selective it is.

Five functional types

TYPE	STRUCTURE	WHERE
1. Exchange	Thin, flattened, simple squamous. Built for rapid diffusion of gases.	Capillaries, lungs. Endothelium is the simple squamous lining of the heart and vessels.
2. Transporting	Cuboidal or columnar cells, microvilli on the apical surface to increase surface area, basolateral folds, tight junctions, many mitochondria. Transport is regulated, for example by hormones.	Intestine, kidney
3. Ciliated	Nontransporting. Cilia move fluid across the surface.	Respiratory tract, female reproductive tract
4. Protective	Stratified layers resisting mechanical and chemical stress, often keratinized, high turnover.	Epidermis, mouth, esophagus, vagina
5. Secretory	Synthesizes and releases products. Single cells or glands.	See the glands below

Glands

- **Exocrine glands** have ducts, and the secretion exits the body. Sweat, saliva, pancreatic enzymes. Serous secretions are watery and enzyme rich, mucous secretions are sticky glycoproteins, and goblet cells produce mucus.
- **Endocrine glands** are ductless. They secrete hormones into the ECF and then into blood. Thyroid, pituitary, gonads. Endocrine cells can also be scattered rather than gathered into a gland.

Retinoids increase turnover in protective epithelia, and smoking paralyzes cilia in the respiratory tract, which raises the

17. Connective tissue

The defining feature is abundant extracellular matrix with cells widely spaced. It provides support, protection and defense.

Matrix structure

- Ground substance is proteoglycans plus water.
- Insoluble fibers are embedded in it.
- The living cells constantly remodel the matrix. It is never finished.

Cell types, by their endings

- **-blast** secretes matrix. Fibroblast, osteoblast.
- **-cyte** maintains it.
- **-clast** breaks it down.
- Fixed cells do maintenance. Mobile cells do defense.

Learn the three endings once and you can read the name of almost any connective tissue cell without being taught it.

Matrix fibers

- **Collagen.** The most abundant protein in the body. Flexible but inelastic, and by weight its tensile strength is greater than steel.
- **Elastin.** Stretch and recoil.
- **Fibrillin.** Forms the scaffolding for elastic fibers.
- **Fibronectin.** Connects cells to the matrix, and is important in wound healing and clotting.

The seven types

1. Loose connective tissue
2. Dense connective tissue. Tendons connect muscle to bone, ligaments connect bone to bone.
3. Adipose tissue. White fat stores energy, brown fat produces heat.
4. Cartilage. Flexible, no blood supply, slow to heal.
5. Bone. Calcified matrix, rigid support.
6. Blood. The matrix is plasma and the cells are suspended in it.
7. Lymph, often grouped with blood.

Predict first

Cartilage has no blood supply. Predict two consequences for a patient with a cartilage injury.

– Check yourself

It heals slowly, because everything a repair needs has to diffuse in rather than arrive by vessel. And drugs given systemically reach it poorly for the same reason. No supply line changes both repair and treatment.

18. Muscle and neural tissue, the excitable tissues

What they share

- They generate action potentials.
- They have minimal extracellular matrix.
- They are often connected by gap junctions.

Muscle tissue

Contraction produces force and movement. Three types: skeletal, which is voluntary; cardiac, in the heart; and smooth, in organs.

Neural tissue

- Neurons transmit signals.
- Glial cells support and protect them.
- Concentrated in the central nervous system but extending body wide.

These two tissues are the whole subject of Weeks 3 through 5. What you need from this page is the structural half: minimal matrix, gap junctions, and the ability to generate an action potential.

19. Tissue remodeling

Necrosis and apoptosis

	NECROSIS	APOPTOSIS
What it is	Uncontrolled cell death	Programmed cell death, tidy
Causes	Trauma, toxins, ischemia	Normal developmental and maintenance signals
Features	Swelling, rupture, inflammation. Released enzymes damage neighboring cells.	Chromatin condensation, cell shrinkage, membrane blebs, then phagocytosis. No inflammation.
Roles	Always pathological	Development, including removing the webs between fingers. Tissue turnover. Intestinal epithelium is replaced every 2 to 5 days.

The inflammation line is the one to hold. Necrotic cells rupture and spill enzymes onto their neighbors, so the damage spreads. Apoptotic cells are packaged up and taken away, so it does not.

Stem cells

- **Totipotent.** The earliest embryonic cells, able to form an entire organism.
- **Pluripotent.** Many cell types, but not an entire organism.
- **Multipotent.** Adult stem cells, restricted lineages. Bone marrow to blood is the example.
- **iPS cells.** Adult cells reprogrammed back to pluripotent. Yamanaka, 2006, Nobel Prize 2012. The advantages are that no embryos are needed and that disease can be modeled in a dish. The challenges are controlling differentiation and avoiding immune rejection.

20. Organs

An organ is a structure made of two or more tissue types that carries out related functions.

Skin as the model organ

- It contains all four tissue types.
- It is the largest organ, roughly 16 percent of body weight.
- Functions: protection, temperature regulation, sensation, and vitamin D synthesis.

Skin is the example to use whenever you are asked what an organ is, because you can name all four tissues in it and four different functions without straining.

Before you move on

Cover this page and draw, from memory, one cell with its membrane, three organelles of your choice, and the nucleus, then draw the route a secreted protein takes from DNA to the extracellular fluid. If any step is missing, you know which section to reread.

Where this goes next

Compartments create barriers. Barriers have to be crossed. That is the transport material, and it is the next thing in Week 2. After that comes signaling, which is how one compartment tells another what to do without anything physically crossing.

Keep this page open while you fill your Competency Study Guide. The competency boxes for the membrane and for signaling are on the same sheet.

Transport across the membrane

The second half of Week 2. The first half gave you the cell as a structure; this half is what that structure does. Diffusion and osmosis, the carriers and the pumps, endocytosis and exocytosis, and how a sheet of cells moves something all the way from the gut into the blood. Silverthorn chapter 5.

COMPETENCIES 5 TO 16 OF WEEK 2

9 SECTIONS · 5 WORKED PROBLEMS

HOW TO USE THIS SHEET

Pass one, before the video: read a section, then close the page and put what you can into the box for that competency on your **Cellular physiology and transport** Competency Study Guide. Pass two, after the video: come back in your second color and add what you missed. Try each worked problem yourself before you read the work. Getting it wrong first is how the method sticks. Printing this page and keeping it beside you works just as well as reading it on screen.

The problem for today

WHERE THIS SITS

The first half of this week gave you the cell as a structure: the membrane, the organelles, the four tissue types, and the junctions that hold a sheet of cells together. Nothing in it moved.

This half is about movement. A cell is a bag of concentrated solution sitting in a different solution, and it stays alive by controlling what crosses the boundary between them. Every mechanism below answers the same two questions: which way is this substance going relative to its own gradient, and is the cell paying for the trip?

Hold those two questions. If you can answer them for any substance at any membrane, you can work out the rest, and most of what looks like memorization in this half turns out to be those two answers in different clothes.

The two questions, and the map they make

THE FRAME FOR EVERYTHING BELOW

Put the two questions on two axes and you get every transport mechanism in physiology.

Down its gradient, no energy spent. This is passive transport. Simple diffusion if the substance crosses the lipid on its own, facilitated diffusion if it needs a protein to get through. Osmosis is this same thing for water.

Against its gradient, energy spent. This is active transport. Primary if the protein burns ATP itself, secondary if it spends a gradient that some other pump built with ATP earlier.

Too big for any protein. The cell wraps it in membrane instead, and that is vesicular transport: endocytosis coming in, exocytosis going out.

When you meet something new, put it on that map first. A student who knows the sodium potassium pump is primary active transport and that SGLT1 is secondary has understood the real difference between them, which is where the ATP was spent.

Simple diffusion, and what sets its speed

COMPETENCY 5

Diffusion is molecules moving from where there are more of them to where there are fewer, driven by nothing but their own random motion. Nobody pushes them. There are simply more ways to end up spread out than stacked in one corner, so over time they spread out.

The important consequence is that diffusion needs no energy and no protein. It also never stops at equilibrium: molecules keep moving in both directions, but the two rates become equal, so the net movement is zero. Equilibrium is a standoff, not a stillness.

Fick's law names the four things that set the rate. Rate goes up with the **concentration gradient**, up with the **surface area** available, down with the **thickness** of the membrane, and down with the **distance** the substance has to travel. Membrane permeability rolls into it too, and that depends on how lipid soluble the substance is, how big it is, and the temperature.

Every one of those four shows up again later as a clinical finding. Emphysema destroys alveolar walls and cuts surface area. Pulmonary edema puts fluid between the air and the blood and increases the distance. Fibrosis thickens the barrier. None of those are new physiology; they are Fick's law with one variable moved.

What crosses the lipid unaided, in order: small nonpolar gases like oxygen and carbon dioxide go easily, small uncharged polar molecules like water and urea go slowly, and large polar molecules and every ion do not go at all without help. That order is worth knowing cold, because it decides which mechanism a substance has to use.

WORKED PROBLEM 1

Oxygen moves from an alveolus into a capillary. A patient develops pulmonary edema, which puts a layer of fluid between the air and the blood and doubles the distance oxygen has to travel. Nothing else changes. What happens to the rate of oxygen transfer, and why does this patient feel short of breath rather than simply tired?

Fick's law: rate rises with gradient and surface area, and falls with thickness and distance.

THE WORK

- 1 Name the variable that changed. Distance went up. Gradient, surface area and membrane thickness are all unchanged.
- 2 Distance sits in the denominator, so doubling it roughly **halves** the rate of oxygen transfer across that patch of membrane.
- 3 Less oxygen crosses per second, so arterial oxygen falls.
- 4 Chemoreceptors respond to that fall by driving ventilation up, which is what the patient experiences as breathlessness.
- 5 The picture to draw: two boxes with a barrier between them, an arrow for oxygen, and the barrier drawn twice, once thin and once with fluid added. Label the distance on both.
- 6 The point to carry: nothing was wrong with the patient's oxygen, their hemoglobin or their gradient. One geometric variable moved and the whole system followed.

hypotonic. The two words are not synonyms and one of them is about the membrane, not the solution.

WORKED PROBLEM 2

A red blood cell is dropped into each of three solutions. A: 300 mOsM sodium chloride. B: 300 mOsM urea. C: 150 mOsM sodium chloride. Sodium chloride does not cross the red cell membrane; urea does. Predict what happens to the cell in each, and give both the osmolarity word and the tonicity word.

Cell interior is about 300 mOsM, and all of it is nonpenetrating.

THE WORK

- 1 Solution A. 300 mOsM, none of it penetrating. Nonpenetrating outside equals nonpenetrating inside, so water has no reason to move. **Isosmotic and isotonic. No volume change.**
- 2 Solution B. 300 mOsM, but urea crosses. Count only the nonpenetrating solute outside: zero. The cell has 300 mOsM of nonpenetrating solute inside and there is none outside, so water moves in.
- 3 Urea also moves in, down its own gradient, which raises the inside osmolarity further and pulls in still more water. The cell swells and lyses. **Isosmotic but hypotonic.**
- 4 Solution C. 150 mOsM, nonpenetrating. Half the nonpenetrating solute outside that the cell has inside, so water moves in and the cell swells. **Hyposmotic and hypotonic.**
- 5 The picture to draw: three cells in three beakers, an arrow for water in each, and under each one write the two words separately so you are forced to notice when they disagree.
- 6 The point to carry: osmolarity is a property of a solution you can measure in a beaker. Tonicity only exists once you name a membrane, because it depends on what that membrane will let through.

Osmosis: the same idea, seen from the water's side

COMPETENCIES 6 AND 7

Osmosis is diffusion of water. Water moves down its own concentration gradient, which means it moves toward the side with **more solute**, because more solute means less water per liter.

Students lose marks here by tracking the solute. Track the water. If one side has more dissolved particles, that side has proportionally less water, and water moves in.

Osmotic pressure is the pressure you would have to apply to stop that movement. It is a way of measuring how badly water wants to move, not a push the solutes are exerting.

Osmolarity is a count: how many osmoles of solute sit in a liter of solution. Body fluids run at about 300 milliosmoles per liter. It counts every particle, and it tells you nothing on its own about what a solution will do to a cell.

Tonicity is the thing that actually matters to a cell, and it counts only the solutes that **cannot cross the membrane**. A solute that crosses simply follows the water and changes nothing in the end.

That distinction is the single most tested idea in this half of the week. A 300 mOsM solution of urea is isosmotic with the cell, and it will still burst a red blood cell, because urea crosses the membrane, enters the cell, and water follows it in. Isosmotic,

Facilitated diffusion: a protein, but still no bill

COMPETENCY 8

Glucose is large and polar. It will not cross a lipid bilayer in any useful amount, and yet every cell in your body takes it up. It goes through a carrier protein, the GLUT family, down its own concentration gradient.

That is facilitated diffusion: protein mediated, still passive, still downhill, still no ATP. The protein is a door, not an engine. Take the gradient away and the movement stops, and if the gradient reverses the protein happily carries glucose the other way.

Two features come with using a protein, and both separate facilitated diffusion from simple diffusion on a graph.

Specificity. A carrier binds a particular shape. GLUT carries glucose and some related sugars and will not carry amino acids.

Saturation. There is a finite number of carriers. Raise the concentration far enough and every carrier is occupied, and the

rate stops rising. Simple diffusion has no such ceiling; its graph is a straight line that keeps climbing.

Channels are the other protein route. A channel is an open pore rather than a binding site, so it moves ions far faster than a carrier moves anything, and it is gated: voltage gated, chemically gated, or mechanically gated. The gating is what Week 3 is built on, so notice it now.

Primary active transport: paying with ATP

COMPETENCY 9

Everything above moves things downhill. A cell that only did that would come to equilibrium with its surroundings, and a cell at equilibrium with its surroundings is dead. Life is the business of holding gradients that would rather collapse.

Primary active transport is a protein that hydrolyzes ATP itself and uses that energy to move a substance against its gradient.

The one to know is the **sodium potassium ATPase**. Per ATP it moves **three sodium ions out** and **two potassium ions in**, both against their gradients. It is in the membrane of every cell you have, and a substantial share of your resting metabolic rate goes to running it.

Two consequences follow, and both matter later.

Because three positive charges leave and only two come in, the pump is **electrogenic**: it makes the inside of the cell slightly more negative. That is a small direct contribution to the resting membrane potential you meet in Week 3.

Because it holds sodium low inside the cell, it creates a steep inward sodium gradient. That gradient is stored energy, and the whole next section is about cells spending it.

Other primary pumps worth naming: the calcium ATPase that keeps intracellular calcium about ten thousand times lower than outside, and the proton pump in the stomach that acidifies gastric juice.

Secondary active transport: spending somebody else's gradient

COMPETENCY 10

A secondary active transporter moves one substance **down** its gradient and uses the energy released to drag a second substance **up** its own. No ATP touches this protein. The ATP was spent earlier, by the sodium potassium pump, building the gradient this transporter now spends.

That is why it is called secondary. It is one step removed from the ATP, and if you poison the sodium potassium pump, every secondary transporter in the cell fails shortly afterwards, because the gradient they all live on runs down.

Two shapes, and the names are literal.

Symport, both substances moving in the same direction. SGLT1 in the intestine and the kidney brings sodium in down its gradient and drags glucose in with it, against the glucose gradient.

Antiport, the two moving in opposite directions. The sodium calcium exchanger lets sodium in and throws calcium out.

Digoxin is the clinical case that ties the two sections together. It blocks the sodium potassium pump, so intracellular sodium rises,

so the sodium gradient the sodium calcium exchanger depends on gets weaker, so less calcium leaves the cell, so intracellular calcium rises and the heart contracts more forcefully. Four steps, no new physiology, and a drug that has been in use for two hundred years.

WORKED PROBLEM 3

A patient with cholera has watery diarrhea and is losing salt and water fast. Intravenous fluid is not available. Oral rehydration solution, which is water with salt and glucose in it, saves the patient's life. Plain salt water does not work nearly as well. Explain why the glucose matters, using the transport proteins involved.

Cholera toxin drives chloride and water secretion into the gut lumen. It does not damage SGLT1 on the apical membrane of the intestinal cell.

THE WORK

- 1 Name the protein the glucose is for. SGLT1 on the apical membrane is a **symport**: sodium in, glucose in, the sodium moving down its gradient and paying for the glucose.
- 2 For SGLT1 to run, there has to be sodium in the lumen and a sodium gradient across the apical membrane. The sodium potassium pump on the basolateral membrane keeps intracellular sodium low, so the gradient is there.
- 3 With glucose present, every glucose molecule absorbed brings a sodium ion in with it. Salt alone gives no such coupling and is absorbed far less efficiently.
- 4 Sodium accumulating inside the cell is pumped out across the basolateral membrane into the interstitial fluid, which raises the osmolarity there.
- 5 Water follows that solute by osmosis, out of the lumen and into the body. The water is never actively transported; it follows the sodium the whole way.
- 6 The picture to draw: one intestinal cell with an apical and a basolateral membrane, SGLT1 on the apical side, the sodium potassium pump on the basolateral side, and a dotted arrow for water following the sodium.
- 7 The point to carry: the treatment works because cholera toxin breaks one pathway and leaves another intact. Knowing which proteins sit on which membrane is what lets you see that.

Transport maximum, and why glucose appears in urine

COMPETENCY 11

Any transport that uses a protein can saturate, because there is a finite number of proteins. The rate at which every carrier is working flat out is the **transport maximum**, the T_m .

Below the T_m , raising the concentration raises the rate. At the T_m , raising the concentration does nothing at all, and whatever cannot be carried goes somewhere else.

The kidney is the clean example. Filtered glucose is normally reabsorbed completely, so there is no glucose in urine. Raise plasma glucose far enough and the carriers saturate, and glucose starts appearing in the urine. The plasma concentration at which that begins is the **renal threshold**, around 180 to 200 mg/dL in most people, and the T_m itself is around 375 mg/min.

This is why uncontrolled diabetes produces sugar in the urine, and why it also produces thirst and high urine volume: glucose left in the tubule holds water there by osmosis, so more water is lost. Saturation of one carrier explains three symptoms.

Simple diffusion never saturates. If you are given a graph that flattens out, a protein is involved somewhere.

Vesicular transport: too big for any protein

COMPETENCY 12

Proteins, bacteria and droplets of extracellular fluid are far too large for any channel or carrier. The cell moves them by wrapping them in a piece of its own membrane.

Phagocytosis, cell eating. The cell sends out actin driven pseudopods that surround a large particle, a bacterium or a dead cell, and pinch off a **phagosome**. The phagosome fuses with a lysosome and the contents are digested. This is not a general cell ability: it belongs to specialized cells, chiefly macrophages and neutrophils, and it is the front line of your immune defense.

Pinocytosis, cell drinking. The membrane dimples inward and pinches off a small vesicle of extracellular fluid with whatever happens to be dissolved in it. It is nonselective and it goes on continuously in most cells.

Receptor mediated endocytosis, the selective one. Receptors on the surface bind a specific ligand, the loaded receptors gather in a region coated on its inner face with **clathrin**, and that coated pit invaginates and pinches off. This is how cells take up cholesterol carried as LDL, and how they take up iron carried on transferrin. Familial hypercholesterolaemia is a defect in the LDL receptor: the cholesterol is in the blood, the cell cannot bring it in, and plasma cholesterol runs very high from childhood.

Exocytosis, the way out. A vesicle moves to the plasma membrane, fuses with it, and empties its contents outside. It does two jobs at once: it secretes the contents, and it adds the vesicle membrane, with whatever proteins were in it, to the plasma membrane. Regulated exocytosis, the kind that releases neurotransmitter or insulin, is triggered by a rise in intracellular calcium, which is worth remembering when you meet synapses in Week 3.

Transcytosis is endocytosis on one face and exocytosis on the other, carrying something straight across a cell. Capillary endothelium moves large proteins this way.

WORKED PROBLEM 4

A child is found to have a plasma cholesterol several times the normal value from a young age, and the family history shows the same pattern. Investigation shows the LDL receptor is defective. Explain why plasma cholesterol is high, and say which step of the transport pathway has failed.

LDL carries cholesterol in the blood. Cells take it up by receptor mediated endocytosis.

THE WORK

- 1 Name the mechanism. Cholesterol arrives as LDL, which is far too large to cross the membrane by any protein route, so it is taken up by **receptor mediated endocytosis**.
- 2 Name the step. The LDL receptor binds LDL at the cell surface and the loaded receptors gather in a clathrin coated pit, which invaginates and pinches off.
- 3 With a defective receptor, LDL is never bound, so no coated pit forms and no vesicle is taken in. The uptake step fails at the very beginning.
- 4 The cholesterol is not missing from the body. It is in the plasma and cannot get out of it, so plasma concentration rises and stays high.
- 5 Cells sense they are short of cholesterol and make more of their own, which makes the plasma level worse rather than better.
- 6 The picture to draw: a cell surface with receptors, LDL particles outside, a coated pit forming in the normal case, and the same drawing with the receptor crossed out and LDL piling up outside.
- 7 The point to carry: a transport defect does not always mean too little of a substance somewhere. Here there is far too much of it, in the wrong compartment, because the door into the right one is broken.

Transepithelial transport: two membranes, two sets of proteins

COMPETENCY 13

Everything so far moved something across one membrane. Absorbing a nutrient from the gut, or reabsorbing it in the kidney, means moving it across a whole **sheet of cells**, in one side and out the other.

That only works because an epithelial cell is **polarized**: the membrane facing the lumen, the **apical** membrane, carries different proteins from the membrane facing the blood, the **basolateral** membrane. The tight junctions from the first half of this week are what keep the two sets from mixing, which is where the anatomy and the physiology of this week meet.

Glucose absorption in the small intestine is the pathway to know, and it uses three proteins on two membranes.

On the **apical** membrane, SGLT1 brings glucose in against its gradient, powered by sodium coming down its gradient. Secondary active transport.

On the **basolateral** membrane, GLUT2 lets glucose leave down its gradient into the interstitial fluid. Facilitated diffusion.

Also on the **basolateral** membrane, the sodium potassium pump throws sodium out, keeping intracellular sodium low so the apical gradient never runs down. Primary active transport.

Three mechanisms, one molecule, one trip. That is the pathway to be able to draw from blank paper, and the question to ask yourself at every step is still the same two: which way relative to the gradient, and who is paying.

There is also a route **between** the cells rather than through them, the paracellular route, and how leaky it is depends on the tight junctions. A leaky epithelium like the proximal tubule lets a good deal through between cells; a tight one like the distal nephron or the bladder lets almost nothing.

WORKED PROBLEM 5

In an experiment, a drug is applied that blocks only the basolateral sodium potassium pump in an intestinal epithelial cell. Glucose absorption across the epithelium falls to almost nothing within minutes, even though the drug never touched SGLT1. Explain the sequence.

SGLT1 on the apical membrane is a sodium glucose symport. GLUT2 on the basolateral membrane is facilitated diffusion.

THE WORK

- 1** Name what the pump was doing. It threw sodium out of the cell across the basolateral membrane, holding intracellular sodium low.
- 2** Block it, and sodium stops leaving. Intracellular sodium begins to rise.
- 3** SGLT1 runs on the sodium gradient across the apical membrane. As intracellular sodium rises, that gradient shrinks.
- 4** With a small gradient there is little energy to spend, so SGLT1 can no longer drag glucose in against the glucose gradient. Apical uptake falls.
- 5** Nothing is wrong with SGLT1 itself. It is a secondary active transporter and its fuel supply was cut off two steps upstream.
- 6** The picture to draw: the same polarized cell as before, with the pump crossed out and a rising sodium level written inside the cell, and an arrow through SGLT1 getting thinner.
- 7** The point to carry: this is what secondary means. Every secondary transporter in the body is one pump failure away from stopping.

together, especially tight junctions and what they do to the route between cells.

From Week 1: the idea of a regulated variable and a control loop. Most of what is being transported here is something a loop somewhere is defending.

From the chemistry review, if you did it: polar and nonpolar, and why a charged particle will not cross an oily layer on its own. If that sentence is not solid, the whole first section rests on it.

Going forward into Week 3: ion channels and their gating, and the sodium potassium pump holding the gradients, are what the resting membrane potential and then the action potential are built from. You have already met both. Week 3 puts a voltage on them.

Close the page and produce it

These are the twelve Week 2 competencies this half of the week covers, straight off the study guide, with the two prompts that go with each one. The other thirteen competencies of Week 2 are the cell and tissues half, and they have their own notes and their own Competency Study Guide. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

01

Simple diffusion and Fick's law

State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer.

Do one of the two. Head your paper with the number.

1-1

Draw the Fick's law setup as a picture, not an equation: two compartments, a membrane between them, and label on the drawing the surface area, the concentration difference, and the membrane thickness. Then draw four small graphs in a row, rate of diffusion against each of those, plus against diffusion distance. Label both axes on every graph. One line: one variable sits in the denominator, so say which and how the shape of its graph shows that.

1-2

Draw a real gas exchange surface, one alveolus against one capillary, and mark on it every Fick variable the body has already optimized. Then draw the same surface in pulmonary edema, with fluid thickening the barrier. Show what changed and what did not. One line: name the single Fick variable the disease changed, and say whether breathing faster fixes it.

What this half assumes you already have

BEFORE YOU GO ON

From the first half of this week: the membrane as a lipid bilayer with proteins in it, and the junctions that hold an epithelium

02

Osmolarity and tonicity

Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell.

Do one of the two. Head your paper with the number.

2-1

Draw three beakers labeled 150 mM NaCl, 300 mM sucrose, and 300 mM urea. Under each, write the osmolarity calculation and the answer in mOsm/L. Then draw a red blood cell in each beaker and show its final volume. One line: all three have the same osmolarity and one of them lyses the cell, so name the property of the solute that decides tonicity.

2-2

Draw three cells that have just come out of three unknown solutions: one shrunken, one swollen but intact, and one lysed. Beside each, write two different solutions that could have produced that result, one differing in osmolarity and one differing only in penetrating solute content. One line: say what osmolarity alone failed to tell you about the lysed cell.

03

Osmosis and cell volume

Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content.

Do one of the two. Head your paper with the number.

3-1

Draw a cell at normal volume in the middle of your page, 290 mOsm/L inside. Around it draw four more cells, in 150 mOsm/L NaCl, 290 mOsm/L NaCl, 450 mOsm/L NaCl, and 290 mOsm/L urea. For each, draw the water arrow with its direction and the final cell volume. One line: for the urea cell, say what happens at the first instant and what happens over the next few minutes.

3-2

Draw one set of axes, cell volume on the y and time on the x, with the moment of immersion marked. Plot three traces on it: a cell in 450 mOsm/L NaCl, a cell in 150 mOsm/L NaCl, and a cell in 600 mOsm/L urea. One line: one trace goes down and then climbs back past where it started, so say which solute movement caused each half of that curve.

04

Diffusion and osmosis experiment

Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient.

Do one of the two. Head your paper with the number.

4-1

Before you run anything, draw your predicted result: the bag or the plate as it will look at the start and at the end, with each test molecule drawn where you expect it to be. Run it, then draw the actual result beside your prediction in a second color. One line: name the substance that moved fastest and say whether molecular size or starting gradient explains it better in your own data.

4-2

Draw a graph of distance moved on the y against molecular weight on the x, one labeled point for every substance you tested. Beside it draw a second small graph, distance moved against starting concentration. One line: your membrane passed some things and stopped others, so write the size cutoff your data actually supports and how confident you are in that number.

05

Tonicity and red blood cells

Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result.

Do one of the two. Head your paper with the number.

5-1

Draw three microscope fields, one per solution: distilled water, 0.9 percent NaCl, and 3 percent NaCl. In each, draw the cells as you actually see them, keeping the biconcave disc, the spiky crenated form, and the ghosts distinct. Under each field write the direction of net water movement. One line: name the slide that is hardest to count cells on, and say why that difficulty is itself the result.

5-2

Draw one red cell in the center of your box and three arrows out to three versions of it: crenated, normal, and hemolysed. Beside each write the tonicity that produced it and the water arrow. Then draw what the membrane itself did in the hemolysed one. One line: crenation is reversible and hemolysis is not, so say what makes the difference.

06

Facilitated diffusion

Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not.

Do one of the two. Head your paper with the number.

6-1

Draw a channel protein and a carrier protein in the same stretch of membrane, with a molecule crossing through each. Make the difference visible in the drawing: the channel as an open pore, the carrier changing shape around its cargo. Underneath, draw one graph of transport rate against solute concentration with two traces, simple diffusion and carrier mediated. One line: only one trace plateaus, so say what has run out at the plateau.

6-2

Draw a carrier protein with glucose bound, then draw three failures beside it: galactose competing for the same carrier, a molecule of the wrong shape bouncing off, and every carrier already occupied. Label which drawing is competition, which is specificity, and which is saturation. One line: adding more glucose helps in one of those three and does nothing in another, so name each and say why.

07

Primary active transport

Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains.

Do one of the two. Head your paper with the number.

7-1

Draw the sodium potassium ATPase in a membrane with inside and outside labeled. Sketch the cycle in three or four steps: ions binding, ATP splitting, the shape change, ions releasing. Write 3 and 2 on the correct arrows in the correct directions. Beside it note the sodium and potassium concentrations inside and out that the pump maintains. One line: it moves three out for every two in, so say what that does to charge across the membrane.

7-2

Draw a cell with the two gradients the pump maintains, sodium high outside and potassium high inside, as tall and short bars on each side. Now poison the pump with ouabain and draw the same bars ten minutes later and an hour later. One line: pick one downstream process that fails, not the resting membrane potential, and say how many steps removed from the pump it is.

08

Secondary active transport

Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger.

Do one of the two. Head your paper with the number.

8-1

Draw a cell membrane with the sodium potassium ATPase keeping sodium low inside. Beside it draw SGLT carrying sodium and glucose in together, and the sodium calcium exchanger sending calcium out as sodium comes in. Draw every ion arrow, and mark the sodium gradient as the shared energy source. Label which one is symport and which is antiport. One line: no ATP touches either of these two proteins, so say where their energy came from.

8-2

Draw the sodium calcium exchanger working normally, then draw the same exchanger in a cell whose sodium potassium pump has been stopped. Use three panels with arrows between them: what happens to intracellular sodium, then to the exchanger's driving force, then to intracellular calcium. One line: this is how digoxin strengthens a failing heart, so write the last step of that chain in your own words.

09

Transport maximum and saturation

Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia.

Do one of the two. Head your paper with the number.

9-1

Draw one set of axes, transport rate on the y and solute concentration on the x. Draw a carrier mediated curve that rises and then flattens, and mark the transport maximum with a labeled dashed line. On the same axes draw a straight simple diffusion line for contrast. One line: past the transport maximum the curve is flat, so say what is happening to every extra solute molecule that arrives.

9-2

Draw a kidney tubule with glucose filtered in, reabsorbed by carriers, and any leftover carried on to urine. Draw it three times: normal plasma glucose, plasma glucose at the renal threshold near 180 mg/dL, and plasma glucose well above that. Show the urine in each. Write the glucose transport maximum, about 375 mg/min, beside them. One line: say why a person can be losing glucose in urine while every carrier is working perfectly.

10

Vesicular transport

Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement.

Do one of the two. Head your paper with the number.

10-1

Draw one cell with four events happening on its surface at once: phagocytosis of a bacterium, pinocytosis of a droplet of fluid, receptor mediated endocytosis of a particle held in a coated pit, and exocytosis of a secretory vesicle. Draw the membrane actually deforming in each. Beside each write the trigger and the cargo. One line: all four cost ATP, so name the one that is completely unselective about what it takes in, and say what that costs the cell.

10-2

Draw one particle's whole journey by receptor mediated endocytosis, left to right across your page: binding to receptors, the coated pit, the vesicle pinching off, fusion with a lysosome, and the receptors returning to the surface. Underneath, draw the reverse journey for a molecule leaving by exocytosis. One line: pick a cell type, not a neuron, and say which direction it spends more energy on.

11

Transepithelial transport

Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane.

Do one of the two. Head your paper with the number.

11-1

Draw one polarized epithelial cell, lumen on the left and blood on the right, tight junctions at the top, apical and basolateral membranes labeled. Trace one glucose molecule all the way through: SGLT at the apical membrane, GLUT at the basolateral membrane, and the sodium potassium pump on the basolateral side. Mark each step active or passive. One line: the pump never touches glucose, so say how it powers the apical step.

11-2

Draw the same epithelium twice. In the first, show what would go wrong if the basolateral pump drifted around to the apical side, and use that to say what tight junctions are for. In the second, draw the cell with apical SGLT blocked. Show where glucose ends up in each. One line: oral rehydration solution contains both salt and sugar, so say which of your two drawings explains why.

12

Transport simulation

Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison.

Do one of the two. Head your paper with the number.

12-1

Before you run it, draw three predicted graphs in a row, transport rate against time, one each for simple diffusion, facilitated diffusion, and active transport. Then draw a fourth showing what a metabolic poison does to whichever of the three it affects. Run the simulation and draw the real curves underneath in a second color. One line: say which transport type the poison left untouched, and why it could not touch it.

12-2

Draw a decision tree with three endpoints: simple diffusion, facilitated diffusion, and active transport. At each branch put one test you can actually run in the simulation, reversing the gradient, raising solute until the rate stops climbing, or adding a metabolic poison. Draw the result that sends you down each branch. One line: name the single test that separates active transport from both of the others in one step.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Compartments, cells and tissues. Values given are typical teaching figures, not laboratory reference ranges. Figures © Pearson Education, Silverthorn Human Physiology 9e, used under the instructor license for this course.

How cells talk to each other

The third part of Week 2. The cell as a structure, then what crosses its membrane, and now how one cell sends a message to another. Signal types, where the receptor sits and why, the G protein cascade, the second messengers, amplification, agonists and antagonists, how a signal is switched off, and what a dose response curve tells you. Silverthorn chapter 6.

COMPETENCIES 17 TO 25 OF WEEK 2

11 SECTIONS · 3 WORKED PROBLEMS

HOW TO USE THIS SHEET

Pass one, before the video: read a section, then close the page and put what you can into the box for that competency on your **Cellular physiology and transport** Competency Study Guide. Pass two, after the video: come back in your second color and add what you missed. Try each worked problem yourself before you read the work. Every pathway in the rest of this course is the five steps in this one wearing different clothes, so it is worth the time now. There is a Print button above if you would rather work from paper.

The problem for today

WHERE THIS SITS

You have the cell as a structure, and you have the ways things cross its membrane. Both of those are about a single cell looking after itself.

This section is about cells talking to each other. Your body has roughly thirty trillion cells and they have to act as one thing. A signal molecule released in one place has to reach the right cells somewhere else, be recognized, be converted into something the inside of the cell can act on, be amplified enough to matter, and then be switched off.

Every signal pathway you will meet for the rest of this course, and there are a great many, is the same five steps in a different costume: **signal, receptor, transduction, response, termination**. Learn the shape once here and the endocrine, nervous and cardiovascular weeks stop being new material and start being examples.

Signal types, and how far they travel

COMPETENCY 17

Signals are classified by how they get to their target, and the classification tells you how fast the response is and how widely it spreads.

Gap junctional. Ions and small molecules pass directly from one cell into the next through connexon pores, with nothing released into the extracellular fluid at all. This is the fastest kind, and you met the junction itself in the first half of this week. It is how cardiac muscle contracts as one unit.

Contact dependent. A membrane molecule on one cell binds a receptor on the cell touching it. Immune cells check each other this way.

Autocrine. The cell releases a signal that acts on itself. Common in immune signaling and in growth control.

Paracrine. Released into the interstitial fluid and acting on neighboring cells, over a short distance, then broken down before it travels far. Histamine in an allergic response works this way.

Neurotransmitter. A neuron releases it into a tiny synaptic cleft onto one target cell. Very short distance, very fast, very precise.

Neurohormone and hormone. Released into the blood and carried everywhere. Slow to arrive, long lasting, and it reaches every cell in the body, so specificity has to come from somewhere else.

That last point is the one worth holding. A hormone in the blood touches every cell you have. What decides which cells respond is not where the signal goes; it is which cells carry a receptor for it.

No receptor, no response. That single idea explains why one hormone can do different things in different tissues and nothing at all in most of them.

Where the receptor sits, and what decides that

COMPETENCY 18

There are only two places a receptor can be, and one property of the signal molecule decides which.

Lipophobic signals, meaning water soluble ones: peptides, proteins, catecholamines like epinephrine. They cannot cross the lipid bilayer, so their receptor has to be **on the cell surface**, with a binding site facing out.

Lipophilic signals, meaning lipid soluble ones: steroid hormones, thyroid hormone. They pass straight through the membrane, so their receptor is **inside the cell**, in the cytosol or the nucleus.

Everything else follows from that one fact, and this is the part students find surprising.

A surface receptor cannot reach into the cell itself, so the message has to be relayed by proteins already sitting inside. Those relays are fast, because the machinery is already built and waiting. The response starts in **seconds**, and it stops soon after the signal leaves.

An intracellular receptor binds its signal and the complex acts on DNA, turning genes on or off. Making new protein takes time. The response starts in **hours** and lasts for hours or days after the signal has gone.

So if a question tells you a hormone is a steroid, you already know the receptor is inside, the mechanism is gene transcription, the onset is slow and the effect is long. You have not memorized that; you worked it out from solubility.

One thing to be careful about. Lipophilic hormones do not dissolve well in blood, which is mostly water, so they travel bound to carrier proteins. Only the free fraction can enter cells, and that is why a total hormone level and a free hormone level are different measurements.

WORKED PROBLEM 1

Two patients are given a drug. Patient A gets a peptide hormone and reports an effect within a couple of minutes that has worn off an hour later. Patient B gets a steroid hormone and feels nothing for several hours, but the effect is still measurable two days later. Explain both time courses from the chemistry of the molecules, without looking up either drug.

Peptides are lipophobic. Steroids are lipophilic.

THE WORK

- 1 Patient A. A peptide cannot cross the membrane, so its receptor is on the cell surface.
- 2 Surface receptors work through proteins that are already in the cell: G proteins, enzymes, second messengers. Nothing has to be built, so the response begins in seconds to minutes.
- 3 Those same second messengers are broken down continuously, so when the hormone leaves the receptor the signal collapses quickly. Short onset, short duration.
- 4 Patient B. A steroid crosses the membrane and binds a receptor inside the cell.
- 5 The receptor complex acts on DNA and changes which proteins the cell makes. Transcription and translation take hours, which is the delay.
- 6 Once made, those proteins persist until they are degraded, so the effect outlives the hormone by a long way. Slow onset, long duration.
- 7 The picture to draw: two cells side by side, one with the signal stopping at the surface and a cascade inside, one with the signal walking straight through to the nucleus. Put a clock on each.
- 8 The point to carry: onset and duration are not facts to memorize per hormone. They fall out of where the receptor is, which falls out of whether the molecule dissolves in lipid.

surface mechanism, milliseconds, and it is how the neuromuscular junction works.

G protein coupled receptors. The receptor activates a G protein, which activates an amplifier enzyme, which makes a second messenger. The biggest family by far, and the target of a large share of all prescription drugs.

Receptor enzymes. The receptor has an enzyme site on its inner face, usually a kinase. The insulin receptor is one.

Integrin receptors. Linked to the cytoskeleton, they transduce mechanical information about what the cell is attached to.

G protein coupled receptors, step by step

COMPETENCY 19, PART TWO

This is the pathway to be able to draw from blank paper, because so much else is a variation on it.

The parts: a receptor that crosses the membrane seven times, a **G protein** sitting on the inner face with three subunits called alpha, beta and gamma, an **amplifier enzyme**, and a **second messenger**.

At rest the alpha subunit holds GDP and the whole thing is quiet.

One. The ligand binds the receptor on the outside and the receptor changes shape.

Two. The receptor's inner face now grips the G protein, and the alpha subunit swaps its GDP for GTP. That swap is the switch turning on.

Three. The activated alpha subunit separates and moves along the inner face of the membrane to the amplifier enzyme.

Four. The amplifier enzyme runs and produces second messenger, a lot of it.

Five. The second messenger activates a protein kinase, which adds phosphate groups to target proteins and changes what they do. Phosphorylation is the cell's general purpose on switch.

Six. The alpha subunit hydrolyzes its own GTP back to GDP and reassembles with beta and gamma. The switch turns itself off, which is why the pathway does not run forever.

Worth noticing: the G protein is a timer as well as a switch. How long the alpha subunit takes to hydrolyze GTP sets how long the signal lasts. Cholera toxin works by locking the alpha subunit so it cannot hydrolyze its GTP; the switch jams on, cAMP stays high, chloride and water pour into the gut, and that is the diarrhea.

Second messengers, and the ones to know

COMPETENCY 20

A second messenger is a small molecule made inside the cell in response to a signal outside it. The hormone is the first messenger and never gets in; the second messenger is what actually carries the news around the cytosol.

Cyclic AMP. The amplifier enzyme is adenylyl cyclase, which converts ATP to cAMP. cAMP activates protein kinase A. Epinephrine acting on a beta receptor runs this pathway.

IP3 and DAG. The amplifier enzyme is phospholipase C, which cuts a membrane phospholipid into two messengers at once. IP3 diffuses to the endoplasmic reticulum and opens calcium channels there; DAG stays in the membrane and activates protein kinase C.

The pathway, feature by feature

COMPETENCY 19, PART ONE

Every surface signal pathway has the same parts, and naming them makes any new pathway readable.

The ligand is the signal molecule. **The receptor** binds it, and binding changes the receptor's shape. That shape change is the whole trick: it is how information gets from outside to inside without the molecule itself ever entering.

Transduction is converting the message into a form the inside of the cell can use. **Amplification** happens along the way. **The response** is what the cell actually does: open a channel, run an enzyme, secrete something, contract, divide.

Four families of surface receptor, and you can sort almost anything into one of them.

Receptor channels. The receptor is itself an ion channel. Ligand binds, channel opens, ions flow. This is the fastest possible

Calcium. The most widely used messenger in the body, and it works because the cell holds cytosolic calcium extremely low, about ten thousand times lower than outside. Opening a calcium channel for a moment therefore produces a large proportional change, and calcium binds calmodulin or troponin and changes what they do. That very low resting level, which the calcium pumps from the transport half of this week maintain, is what makes calcium usable as a signal at all.

Cyclic GMP and nitric oxide. Nitric oxide is a gas that diffuses straight into the neighboring cell, raises cGMP, and relaxes smooth muscle. This is how nitroglycerin relieves angina and how sildenafil works.

Amplification: why one molecule matters

COMPETENCY 22

A hormone is present in blood at very low concentration, often a billionth of a mole per liter. The response it produces can be enormous. The gap between those two is amplification, and it is a property of the cascade, not of the hormone.

Each step multiplies. One ligand activates one receptor, but that receptor activates many G proteins before the ligand lets go. Each G protein switches on an amplifier enzyme, and each enzyme makes many molecules of second messenger. Each second messenger activates a kinase, and each kinase phosphorylates many target proteins.

Multiply four steps of roughly ten to a hundred each and one hormone molecule ends up changing millions of product molecules. The classic worked figure for epinephrine driving glycogen breakdown in a liver cell is a single hormone molecule producing on the order of a hundred million molecules of glucose.

Two consequences. Sensitivity: a cell can respond to a vanishingly small signal, which is what lets hormones work at the concentrations they do. And fragility: a cascade with that much gain has to be shut off precisely, or a trace of signal would run away. Which is the next section.

WORKED PROBLEM 2

Epinephrine is released during a fright and reaches the liver at a concentration of about one nanomole per liter, while glucose leaves the liver in millimolar amounts. Work through the cascade and say where the gain comes from, then say what would happen if the G protein could not hydrolyze its GTP.

Epinephrine is lipophobic and acts at a beta receptor. Adenylyl cyclase, cAMP and protein kinase A are involved.

THE WORK

- 1 Epinephrine cannot enter the cell, so it binds a surface receptor. Shape change.
- 2 The receptor activates G proteins. One occupied receptor turns on many G proteins while the ligand is bound, so the first multiplication happens here.
- 3 Each alpha subunit activates adenylyl cyclase, and each enzyme molecule converts many ATP into cAMP. Second multiplication, and the biggest one.
- 4 cAMP activates protein kinase A, and each kinase phosphorylates many targets. Third multiplication.
- 5 The phosphorylated enzymes break glycogen down to glucose, and each of those enzymes works on many glycogen molecules. Fourth.
- 6 Four multiplying steps take one nanomolar signal to a millimolar output. No single step is remarkable; the product of four is.
- 7 If the alpha subunit could not hydrolyze its GTP, the switch would never turn itself off. cAMP would stay high, kinase A would stay active, and the liver would keep dumping glucose with no signal present. That is not hypothetical; it is the mechanism of cholera toxin in the gut and of certain tumors in endocrine tissue.
- 8 The picture to draw: a funnel widening at each step, with the step names on it, and a separate small drawing of the alpha subunit with its GTP crossed out.

Receptor enzymes and intracellular receptors

COMPETENCY 21

Receptor enzymes skip the G protein. The receptor itself carries an enzyme site on its inner face, almost always a tyrosine kinase, and ligand binding switches that enzyme on directly.

The insulin receptor is the one to know. Insulin binds, the receptor autophosphorylates, and a chain of intracellular proteins is phosphorylated in turn. One of the results is that vesicles carrying GLUT4 transporters fuse with the plasma membrane, which raises glucose uptake in muscle and fat. Notice that the final step is exocytosis inserting a facilitated diffusion carrier: three separate ideas from this week in one pathway.

Intracellular receptors are the lipophilic route. The hormone crosses the membrane, binds its receptor in the cytosol or

nucleus, and the complex binds DNA at a response element and changes transcription of specific genes.

Cortisol and the sex steroids work this way, and so does thyroid hormone, which is not a steroid but is lipophilic and behaves like one for this purpose. Slow, long lasting, and it changes what the cell is rather than just what it is doing at this moment.

Turning the volume up and down

COMPETENCY 23

A cell is not a fixed listener. It changes how loudly it hears a signal by changing its receptors, and this is as important clinically as the signal itself.

Down regulation. Sustained high signal, and the cell removes receptors from its surface or makes fewer of them. The same amount of hormone now produces less effect. This is the mechanism behind drug tolerance, and it is part of why a constant infusion of a hormone works less well than a pulsed one.

Up regulation. Sustained low signal, and the cell adds receptors, becoming more sensitive. A denervated muscle does this and becomes hypersensitive to acetylcholine.

Then the two words for molecules that bind receptors.

An **agonist** binds and activates. It does what the natural ligand does. A **full agonist** can produce the maximum response; a **partial agonist** activates the receptor but cannot reach the maximum however much you give.

An **antagonist** binds and does not activate. It occupies the site and blocks the real ligand. A beta blocker is an antagonist at the receptor epinephrine uses.

Competitive antagonists bind the same site and can be outcompeted by enough agonist, so the dose response curve shifts right but still reaches the same maximum. A **noncompetitive** antagonist cannot be outcompeted, so the maximum itself comes down. Which of those two a drug is decides whether more agonist can rescue the patient, which is why the distinction is on the exam and in the clinic.

Signal termination, and what happens when it fails

COMPETENCY 24

Nothing above matters unless the signal can be stopped. A pathway with that much amplification and no off switch is a pathology, not a control system.

The ways a signal ends.

The ligand leaves or is destroyed. Acetylcholinesterase breaks down acetylcholine in the synaptic cleft in about a millisecond. Neurotransmitters are also taken back up into the cell that released them. Hormones are cleared by the liver and the kidney, and a hormone's half life tells you how long its effect persists.

The receptor stops responding. Receptors are internalized or chemically modified so they no longer couple to their G protein.

The switch turns itself off. The G protein hydrolyzes its GTP, as above.

The second messenger is destroyed. Phosphodiesterase breaks cAMP down continuously. Calcium is pumped back out of the cytosol and into the endoplasmic reticulum.

The phosphate comes off. Phosphatases reverse what the kinases did. Every on switch in the cell has a matching off switch, and the state of a protein at any moment is the balance between the two.

Failure of termination is a recognizable category of disease. Cholera toxin jams the G protein on. Some tumors carry a mutation that leaves a receptor permanently active with no ligand at all. Conversely, sildenafil works by **inhibiting termination**: it blocks the phosphodiesterase that would break down cGMP, so the relaxing signal lasts longer. Knowing how a pathway is switched off tells you where a drug can act.

Dose and response, and what a curve tells you

COMPETENCY 25

Plot the response against the dose and the curve is not a straight line. It rises slowly at low doses, steeply in the middle, then flattens.

The flattening is **saturation**, and it is the same idea as the transport maximum from the other half of this week: a finite number of receptors, all occupied. Past that point more drug adds nothing and side effects are all that is left.

Plot dose on a logarithmic axis and the same data becomes a tidy S shape, which is why nearly every published dose response curve is drawn that way.

Two numbers come off the curve.

EC50 is the dose giving half the maximum response, and it measures **potency**: how much drug you need. A lower EC50 means a more potent drug.

Maximal response measures **efficacy**: how much the drug can do at its best.

They are independent, and confusing them is the commonest error here. A very potent drug with a low ceiling is worse for a severe problem than a less potent drug with a high one. A partial agonist has lower efficacy no matter how potent it is.

On the curve: a competitive antagonist shifts it right, so the EC50 rises but the maximum is unchanged. A noncompetitive antagonist lowers the maximum. Being able to look at a shifted curve and say which kind of blocker was added is the competency.

WORKED PROBLEM 3

Two dose response curves are drawn for the same agonist. In experiment one a blocker is added and the curve shifts to the right with the same maximum. In experiment two a different blocker is added and the maximum falls, with no rightward shift. Identify each blocker, and say which patient you could rescue by giving more agonist.

Competitive antagonists bind the same site as the agonist. Noncompetitive ones do not.

THE WORK

- 1 Experiment one. The maximum is unchanged, so every receptor can still produce a full response if the agonist reaches it. More agonist restores the original effect.
- 2 That is a **competitive antagonist**. It binds the same site and the two molecules compete, so raising agonist concentration wins the competition.
- 3 The shift right means a higher EC50: you need more drug for the same effect. Potency has fallen, efficacy has not.
- 4 Experiment two. The maximum is lower, so some receptors cannot produce a response at any agonist concentration.
- 5 That is a **noncompetitive antagonist**. Raising the agonist does not help, because the problem is not competition for the binding site.
- 6 Efficacy has fallen. EC50 may be unchanged.
- 7 Clinically: you can out dose the competitive blocker and you cannot out dose the noncompetitive one. That is the difference between an overdose you can treat with more agonist and one you have to support until the drug clears.
- 8 The picture to draw: one set of axes with three curves on it, the control, the right shifted one, and the flattened one, with EC50 marked on each.

Close the page and produce it

The nine signaling competencies this lecture covers, plus the two from earlier in the week that the other notes name but never teach: determinants of permeability, and compartment separation with its clinical volume shifts. Both belong here, because permeability is what decides where a receptor has to sit, and a volume shift is what a signal is usually defending. Pick one prompt per competency. Blank paper, nothing open.

Read the full instructions first, including how the first pass before the lecture differs from the second pass after it.

01

Signal types and range

Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

Do one of the two. Head your paper with the number.

1-1

Draw one target cell in the middle of the box. Draw an arrow in from each of the five signal types, autocrine, paracrine, neurotransmitter, neurohormone, hormone. Label each arrow with how far it traveled and what carried it.

1-2

Make a five row table. Signal type, where it is released, how far it goes, what it travels through. Fill it from memory, then check one row against the book.

02

Receptor location and ligand solubility

Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

Do one of the two. Head your paper with the number.

2-1

Draw a cell with one surface receptor and one intracellular receptor. Show the path a lipid soluble signal takes and the path a water soluble one takes. Mark which response is faster and which lasts longer.

2-2

Write the two questions you would ask about any new signal molecule to predict where its receptor sits. Then answer both for a steroid and for a peptide hormone.

What this half assumes you already have

BEFORE YOU GO ON

From the cell and tissues half: the membrane with proteins in it, and gap junctions, which are the one signaling route that needs no receptor at all.

From the transport half: ion channels and gating, since a receptor channel is both; the calcium pumps, because the low resting calcium they maintain is what makes calcium usable as a messenger; and exocytosis, because that is how insulin puts GLUT4 into the membrane and how a neuron releases transmitter.

From Week 1: the control loop. A signal pathway is the efferent limb of one, drawn at molecular scale.

Going forward: Week 3 is membrane potential, neurons and synapses, and a synapse is a paracrine signal across a very small gap onto a receptor channel. You have already met every part of it.

03

G protein coupled receptors

Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response.

Do one of the two. Head your paper with the number.

3-1

Draw the membrane with a receptor crossing it, and trace the pathway in numbered order: ligand binds, the G protein swaps GDP for GTP, the alpha subunit leaves, it switches on the amplifier enzyme, the enzyme makes second messenger, the messenger switches on a kinase, the kinase changes a target protein. Beside each step write roughly how many molecules that one step produced from one. One line: name the step where the biggest multiplication happened.

3-2

Draw a Gs pathway and a Gq pathway side by side, starting from the same membrane and ending at two different cellular responses. Label the amplifier enzyme and the second messenger on each. Then draw the off switch on both, GTP being hydrolyzed on the alpha subunit. One line: cholera toxin jams that off switch open, so write what the gut cell keeps doing and why the person loses fluid.

04

Second messengers

Identify cyclic AMP, IP3, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates.

Do one of the two. Head your paper with the number.

4-1

Draw a membrane and put all four second messengers on one page: cyclic AMP, IP3, diacylglycerol, and calcium. For each, draw where it comes from, the enzyme or channel that produced it, and an arrow to one thing it switches on. Keep DAG in the membrane and IP3 out in the cytosol, because that is where they actually go. One line: one of the four is not made by an enzyme at all, so name it and say where it comes from instead.

4-2

Draw phospholipase C cutting PIP2 in the membrane and follow both halves. Send IP3 to the endoplasmic reticulum and draw what comes out. Leave DAG where it is and draw what it activates. Show the two halves converging on one response. Draw the adenylyl cyclase and cyclic AMP route beside it for contrast. One line: cyclic AMP is destroyed by phosphodiesterase, so say what a drug that blocks that enzyme does to the signal.

05

Catalytic receptors and intracellular receptors

Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response.

Do one of the two. Head your paper with the number.

5-1

Draw one cell with the insulin receptor in the membrane on one side and a steroid receptor sitting inside the cell on the other. Trace both from hormone arrival to final response, the steroid crossing the membrane and reaching DNA, the insulin receptor phosphorylating itself and moving a glucose transporter to the surface. Put a clock beside each showing how long the response takes. One line: name the one you could not use if you needed the effect ten seconds from now.

5-2

Draw one time axis with two response curves on it, a receptor enzyme response and a steroid response, each from hormone arrival through peak to fade. Make the onsets and durations honestly different. Under the axis, draw what is physically being changed in each case, an existing protein being modified or a new protein being built. One line: say which of the two you would want for the next thirty seconds and which for the next week.

06

Signal amplification

Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

Do one of the two. Head your paper with the number.

6-1

Draw a three step cascade as a widening funnel. Put a number on each step and multiply them down the side to get the total amplification.

6-2

Start from one ligand and write the count at each step of a cascade until you reach the response. One line per step, numbers only, no sentences.

07

Receptor modulation

Define agonist, antagonist, competitive inhibition, up regulation, and down regulation and predict the effect of chronic ligand excess on target cell sensitivity.

Do one of the two. Head your paper with the number.

7-1

Draw one target cell three times. Normal, after a long stretch of scarce signal, after a long stretch of excess signal. Show the receptor count changing in each and label which is up regulation and which is down.

7-2

Define agonist, antagonist, up regulation and down regulation in your own words, one line each. Then explain in two lines why a patient who needs more opioid for the same effect is usually showing down regulation rather than drug seeking.

08

Signal termination

Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

Do one of the two. Head your paper with the number.

8-1

Draw one signal running from receptor to response across a cell, then add three different off switches to that same drawing at three different points: the ligand removed or destroyed outside the cell, the second messenger broken down inside, and the response protein having its phosphate taken off. Name the enzyme doing each. One line: pick the switch nearest the response and say what still goes wrong if only that one fails.

8-2

Pick a signal, not insulin, and draw it working normally, then draw it again with its termination broken. In the broken version show what piles up and where. Underneath, draw the effect on the tissue as two panels, one over minutes and one over weeks. One line: a signal that never shuts off eventually produces less response rather than more, so write what the cell did to itself.

09

Dose response relationships

Plot a dose response curve, identify threshold, maximal response, and EC50, and compare a full agonist with a partial agonist on the same axes.

Do one of the two. Head your paper with the number.

9-1

Draw a dose response curve with log dose on the x axis and response as a percentage of maximum on the y. Mark threshold, EC50, and the maximal response, with construction lines dropped to the axes. Then plot a partial agonist on the same axes. One line: the partial agonist never reaches the top no matter how much you add, so say what that tells you about the drug that raising the dose cannot fix.

9-2

Draw three curves on one set of axes: a full agonist alone, the same agonist with a competitive antagonist added, and the same agonist with an antagonist that will not come off the receptor. Show which curve slides sideways and which one gets shorter. One line: only one of the two antagonists can be beaten by giving more agonist, so name it and say what feature of the graph proves it.

10

Determinants of permeability

Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein.

Do one of the two. Head your paper with the number.

10-1

Draw a bilayer as a vertical wall down the middle of your page. Line up six molecules on the left: oxygen, carbon dioxide, water, a steroid hormone, glucose, and a sodium ion. Draw an arrow straight through the wall for each one that crosses unaided, a blocked arrow for each that cannot, and a protein set into the wall for each that needs help. One line: sodium is tiny and still cannot cross, so say what actually stopped it.

10-2

Draw a ranking ladder, most permeable at the top and least at the bottom, and place at least six molecules of your own choosing on it. Beside each rung write the properties that put it there: size, lipid solubility, and charge where it applies. One line: name the pair on your ladder that are close in size but far apart in rank, and say what separates them.

11

Compartment separation and clinical volume shifts

Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.

Do one of the two. Head your paper with the number.

11-1

Draw a patient with swollen ankles and a low blood pressure. Use arrows to show where the water went.

11-2

Draw three scenarios: water gained in the vessels, water lost from the body, water moved into tissue. Predict blood pressure under each.

Dr. Sharilyn Rennie · BIO 005 Human Physiology · Compartments, cells and tissues. Values given are typical teaching figures, not laboratory reference ranges. Figures © Pearson Education, Silverthorn Human Physiology 9e, used under the instructor license for this course.